



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	257	Media
HackerNews	3	Community
The Verge AI	1	Media
MIT Tech Review	1	Media

VERIFIED INTELLIGENCE — 3 stories

VERI

SODE: Analyzing Social Dynamics in LLM Agents

ArXiv CS.AI

+2 sources

100%

arXiv:2605.23949v1 Announce Type: cross Abstract: As Large Language Models (LLMs) evolve into interactive agents, understanding their behavioral alignment within human social dynamics becomes essential. While behavioral game theory offers a framework to study these interactions,...

<https://arxiv.org/abs/2605.23949>

VERI

Show HN: skills-for-humanity - 171 structured reasoning skills for Claude Code

HackerNews

+1 source

90%

HN 3

arXiv:2605.04906v2 Announce Type: replace Abstract: While Large Language Models (LLMs) excel in certain reasoning tasks, they struggle in multi-agent games where the final outcome depends on the joint strategies of all agents. In multi-agent games, the non-stationarity of other...

<https://github.com/human-avatar/skills-for-humanity>

VERI

Show HN: CRED-1 - Open domain credibility dataset for on-device pre-bunking

HackerNews

+1 source

90%

HN 3

arXiv:2605.24058v1 Announce Type: cross Abstract: On-device adaptation of large language models commonly keeps a quantized base model frozen while training and deploying a small, task-specific LoRA adapter. In the unmerged adapter-mode setting, however, the adapter is more than...

<https://github.com/aloth/cred-1>

CONFIRMED SIGNALS — 255 stories

CONF

Show HN: Play nbsdgames to increase your brain (exclamation mark)

HackerNews

50%

HN 6

Hello folks, Over the past months I found it extremely hard to find photos for my blog that looked like MY brand. I was tired of using stock photos and also did not want to spend time on canva creating graphics. The AI images also looked very off-brand and I always had to copy...

<https://github.com/abakh/nbsdgames>

CONF

In Search of the Ingredients of Open-Endedness: Replicating Picbreeder with Large Vision-Language Models

ArXiv CS.AI

50%

arXiv:2605.23908v1 Announce Type: new Abstract: We are in the midst of large-scale industrial and academic efforts to automate the processes of scientific, technological and creative production through AI-driven assistants. Historically, a fundamental property of these processes...

<https://arxiv.org/abs/2605.23908>

CONF

How Much Thinking is Enough? Quantifying and Understanding Redundancy in LLM Reasoning

ArXiv CS.AI

50%

arXiv:2605.23926v1 Announce Type: new Abstract: Reasoning-capable large language models solve hard problems by emitting long chains of thought, paying heavily in latency, GPU time, and energy. Casual inspection of their traces reveals extensive reformulation, verification, and...

<https://arxiv.org/abs/2605.23926>

CONF

Toward Reliable Design of LLM-Enabled Agentic Workflows: Optimizing Latency-Reliability-Cost Tradeoffs

ArXiv CS.AI 50%

arXiv:2605.23929v1 Announce Type: new Abstract: Modern AI systems increasingly rely on workflows composed of multiple interacting agents, some powered by large language models (LLMs) and others by conventional computational modules. This paper analyzes the fundamental tradeoffs...

<https://arxiv.org/abs/2605.23929>

CONF

BODHI: Precise OS Kernel Specification Inference

ArXiv CS.AI 50%

arXiv:2605.23931v1 Announce Type: new Abstract: The formal verification of operating system kernels requires precise specifications that capture the intended behavior of system calls. Writing these specifications manually demands deep domain expertise, motivating the use of...

<https://arxiv.org/abs/2605.23931>

CONF

When Correct Beliefs Collapse: Epistemic Resilience of LLMs under Clinical Pressure

ArXiv CS.AI 50%

arXiv:2605.23932v1 Announce Type: new Abstract: Despite strong medical benchmark accuracy, LLMs can exhibit severe multi-turn sycophancy in clinical dialogue, abandoning initial correct diagnosis under escalating pressure. We propose Med-Stress , a targeted...

<https://arxiv.org/abs/2605.23932>

0 CONF

Insuring Every Action: An Authority Frontier Framework for Runtime Actuarial Control of Autonomous AI Agents

ArXiv CS.AI 50%

arXiv:2605.25632v1 Announce Type: new Abstract: Autonomous AI agents increasingly issue side-effect-bearing actions: database mutations, refunds, payments, external commitments. We propose the Actuarial Action Interface (AAI), a deterministic runtime contract that prices each...

<https://arxiv.org/abs/2605.25632>

1 CONF

Fuzzy, Neutrosophic, and Uncertain Graph Theory: Properties and Applications

ArXiv CS.AI 50%

arXiv:2605.23936v1 Announce Type: new Abstract: This book presents a comprehensive and systematic survey of graph theory under uncertainty, with particular emphasis on the unifying role of the uncertain graph framework. It reviews fundamental concepts, structural properties,...

<https://arxiv.org/abs/2605.23936>

2 CONF

Generative OOD-regularized Model-based Policy Optimization

ArXiv CS.AI 50%

arXiv:2605.24405v1 Announce Type: cross Abstract: We study sequential decision-making with offline reinforcement learning (RL). Traditional offline RL policies may result in out-of-distribution (OOD) actions when training relies only on sparse offline representations. To ensure...

<https://arxiv.org/abs/2605.24405>

3 CONF

Authority Inversion in LLM-Mediated Ubiquitous Systems: When Models Trust Users Over Sensors

ArXiv CS.AI 50%

arXiv:2605.23938v1 Announce Type: new Abstract: Large language models (LLMs) increasingly fuse heterogeneous inputs in ubiquitous systems. Yet, how LLMs implicitly allocate authority when sensor measurements and user claims conflict remains unexamined, raising critical...

<https://arxiv.org/abs/2605.23938>

4 CONF

Residual Drift Dominates Contradiction in Multi-Turn Constraint Reasoning

ArXiv CS.AI 50%

arXiv:2605.23940v1 Announce Type: new Abstract: How do multi-turn reasoning systems fail? The expected answer is logical contradiction, in which the system's maintained state becomes unsatisfiable. We show that the dominant mode is instead satisfiable drift, where the internal...

<https://arxiv.org/abs/2605.23940>

5 CONF

MEMOR-E: In-Context and Fine-Tuned LLM Personalization for Alzheimer's Assistive Robotics

ArXiv CS.AI 50%

arXiv:2605.23941v1 Announce Type: new Abstract: Alzheimer's disease is a neurodegenerative disorder marked by progressive declines in memory and language that reduce independence in daily life, motivating socially assistive robotic support. This paper presents MEMOR-E, a mobile...

<https://arxiv.org/abs/2605.23941>

6 CONF

Right-Sizing Communication and Recommendation Set Size in AI-Assisted Search

ArXiv CS.AI 50%

arXiv:2605.23944v1 Announce Type: new Abstract: We model the interaction between a user and an AI driven recommendation system. The user initiates the process by conveying preference information through a costly and noisy message. The AI assistant, acting as a Bayesian agent,...

<https://arxiv.org/abs/2605.23944>

7 CONF

Accelerating Long-Tail Generation in Synchronous RLHF Training via Adaptive Tensor Parallelism

ArXiv CS.AI 50%

arXiv:2605.23945v1 Announce Type: new Abstract: Reinforcement Learning from Human Feedback (RLHF) has become a key post-training paradigm for improving model quality. However, the synchronous three-stage RLHF pipeline is often bottlenecked by the generation stage, where...

<https://arxiv.org/abs/2605.23945>

8 CONF

Methods for Formal Verification of Agent Skills: Three Layers Toward a Mechanically Checkable Capability-Containment Proof

ArXiv CS.AI 50%

arXiv:2605.23951v1 Announce Type: new Abstract: The companion paper introduced a four-level verification lattice on agent-skill manifests (unverified, declared, tested, formal) and left the top level aspirational. This paper closes that gap. We give a precise semantics for skill...

<https://arxiv.org/abs/2605.23951>

9 CONF

Machine Psychometrics: A Mathematical Psychology of Artificial Intelligence

ArXiv CS.AI 50%

arXiv:2605.23952v1 Announce Type: new Abstract: Artificial agents now generate behavior rich enough to invite trust, surprise, and concern, yet our evaluation tools still privilege capability scores over psychological structure. This paper argues that the philosophical impasse...

<https://arxiv.org/abs/2605.23952>

0 CONF

Committed SAE-Feature Traces for Audited-Session Substitution Detection in Hosted LLMs

ArXiv CS.AI 50%

arXiv:2604.18179v2 Announce Type: replace-cross Abstract: Hosted-LLM providers have a silent-substitution incentive: advertise a stronger model while serving cheaper replies. Probe-after-return schemes such as SVIP leave a parallel-serve side-channel, since a dishonest provider...

<https://arxiv.org/abs/2604.18179>

1 CONF

QUIVER: A Formal Framework for Quantifying Perturbation Propagation and Bifurcation in Compound AI Systems

ArXiv CS.AI 50%

arXiv:2605.23956v1 Announce Type: new Abstract: Compound AI systems that chain multiple LLM calls into directed computation graphs are now the dominant architecture for production AI. Although these architectures leverage heterogeneous nodes with mixed-mode outputs, no existing...

<https://arxiv.org/abs/2605.23956>

2 CONF

OSDTW: Optimal Shared Depth and Task Weighting for Long-Tailed Recognition

ArXiv CS.AI 50%

arXiv:2605.24969v1 Announce Type: cross Abstract: Long-tailed recognition suffers from a persistent head--tail trade-off: improving tail performance often degrades head accuracy and can increase training instability. Despite strong empirical results from re-weighting, decoupled...

<https://arxiv.org/abs/2605.24969>

3 CONF

Why We Need World Models for AGI: Where LLMs Fail and How World Models May Outperform

ArXiv CS.AI 50%

arXiv:2605.23972v1 Announce Type: new Abstract: Large language models achieve strong performance in language generation and knowledge-intensive tasks, yet remain limited in settings requiring causal reasoning, persistent state tracking, and long-horizon planning. We argue that...

<https://arxiv.org/abs/2605.23972>

4 CONF

Bridging Evolutionary Algorithms and Reinforcement Learning: A Comprehensive Survey on Hybrid Algorithms

ArXiv CS.AI 50%

arXiv:2401.11963v5 Announce Type: replace-cross Abstract: Evolutionary Reinforcement Learning (ERL), which integrates Evolutionary Algorithms (EAs) and Reinforcement Learning (RL) for optimization, has demonstrated remarkable performance advancements. By fusing both approaches,...

<https://arxiv.org/abs/2401.11963>

5 CONF

Reason--Imagine--Act: Closed-Loop LLM Decision Making with World Models for Autonomous Driving

ArXiv CS.AI 50%

arXiv:2605.24004v1 Announce Type: new Abstract: Large language models (LLMs) are promising for autonomous driving, but semantics-only decision policies can yield physically unsafe behavior in dynamic traffic. Existing methods either perform online language reasoning without...

<https://arxiv.org/abs/2605.24004>

6 CONF

PALoRA: Projection-Adaptive LoRA for Preserving Reasoning in Large Language Models

ArXiv CS.AI 50%

arXiv:2605.24549v1 Announce Type: new Abstract: Efficiently updating Large Language Models (LLMs) with new or evolving factual knowledge remains a central challenge, as even parameter-efficient adaptation can erode previously acquired reasoning abilities. This tension reflects a...

<https://arxiv.org/abs/2605.24549>

7 CONF

SkillEvolBench: Benchmarking the Evolution from Episodic Experience to Procedural Skills

ArXiv CS.AI 50%

arXiv:2605.24117v1 Announce Type: new Abstract: Large language model (LLM) agents accumulate rich episodic trajectories while solving real-world tasks, but it remains unclear whether such experience can be distilled into reusable procedural skills. We introduce SkillEvolBench, a...

<https://arxiv.org/abs/2605.24117>

8 CONF

Benchmarking and Improving Monitors for Out-Of-Distribution Alignment Failure in LLMs

ArXiv CS.AI 50%

arXiv:2605.21602v2 Announce Type: replace Abstract: Many safety and alignment failures of large language models (LLMs) occur due to out-of-distribution (OOD) situations: unusual prompt or response patterns that are unforeseen by model developers. We systematically study whether...

<https://arxiv.org/abs/2605.21602>

9 CONF

Inference Time Context Sparsity: Illusion or Opportunity?

ArXiv CS.AI 50%

arXiv:2605.24168v1 Announce Type: new Abstract: Sparsity has long been a central theme in LLM efficiency, but its role in context processing remains unresolved. As LLM workloads shift toward longer contexts and agentic interactions, the compute and memory bottlenecks of...

<https://arxiv.org/abs/2605.24168>

0 CONF

EPPC-OASIS: Ontology-Aware Adaptation and Structured Inference Refinement for Electronic Patient-Provider Communication Mining in Secure Messages

ArXiv CS.AI 50%

arXiv:2605.24172v1 Announce Type: new Abstract: Secure patient-provider messages contain clinically important communication behaviors that are difficult to characterize manually at scale. The Electronic Patient-Provider Communication (EPPC) framework provides an ontology for...

<https://arxiv.org/abs/2605.24172>

1 CONF

A Sober Look at Agentic Misalignment in Automated Workflows

ArXiv CS.AI 50%

arXiv:2605.24197v1 Announce Type: new Abstract: We study a class of emergent misalignment in multi-agent systems (MAS), with a focus on automated workflows, which we refer to as agentic misalignment. Although these systems can solve complex tasks, they often fail because agents act...

<https://arxiv.org/abs/2605.24197>

2 CONF

When Does Multi-Agent RL Improve LLM Workflows? Workflow, Scale, and Policy-Sharing Tradeoffs

ArXiv CS.AI 50%

arXiv:2605.24202v1 Announce Type: new Abstract: Multi-agent LLM workflows route inference through specialized roles to lift end-task accuracy, but jointly training those roles with reinforcement learning is unstable in ways that are poorly understood. We study when end-to-end RL...

<https://arxiv.org/abs/2605.24202>

3 CONF

Rethinking Weak Supervision in Anomaly Detection: A Comprehensive Benchmark

ArXiv CS.AI 50%

arXiv:2605.26068v1 Announce Type: cross Abstract: Weakly supervised anomaly detection (WSAD) has developed in three primary directions: incomplete, inexact, and inaccurate supervision. However, these directions remain isolated, lacking a unified framework to assess whether they...

<https://arxiv.org/abs/2605.26068>

4 CONF

Human-AI Collaboration in Science at Scale: A Global Large-scale Randomized Field Experiment

ArXiv CS.AI 50%

arXiv:2605.24180v1 Announce Type: cross Abstract: Collaboration is the defining mode of modern science, yet its core mechanism -- feedback -- remains hard to observe, difficult to scale, and unequally distributed. Here we test whether large language models (LLMs) can contribute...

<https://arxiv.org/abs/2605.24180>

5 CONF

ClawTrace: Cost-Aware Tracing for LLM Agent Skill Distillation

ArXiv CS.AI 50%

arXiv:2604.23853v2 Announce Type: replace Abstract: Skill-distillation pipelines learn reusable rules from LLM agent trajectories, but they lack a key signal: how much each step costs. Without per-step cost, a pipeline cannot distinguish adding a missing step to fix a bug from...

<https://arxiv.org/abs/2604.23853>

6 CONF

Hypothesis Generation and Inductive Inference in Children and Language Models

ArXiv CS.AI 50%

arXiv:2605.24528v1 Announce Type: new Abstract: Real world decision-making requires constructing mental models under uncertainty over evidence, over the underlying causal rules, and over the state of the world itself. Which computational principles underpin human inference under...

<https://arxiv.org/abs/2605.24528>

7 **CONF****A governance horizon for ethical-use constraints in open-weight AI models**

ArXiv CS.AI 50%

arXiv:2605.24383v1 Announce Type: new Abstract: Ethical constraints on open-weight AI models are both a reflection of societal concerns and a foundation for AI governance policy. They are expected to propagate to downstream derivatives while implemented as voluntary metadata...

<https://arxiv.org/abs/2605.24383>
8 **CONF****ConceptM³E: Concept-Guided Multimodal Mixture of Experts for Interpretable Computational Pathology**

ArXiv CS.AI 50%

arXiv:2605.24399v1 Announce Type: new Abstract: Healthcare models are transitioning from unimodal prediction toward multimodal reasoning over heterogeneous diagnostic inputs. In computational pathology, for complex tumor subtypes where morphology alone can be challenging to...

<https://arxiv.org/abs/2605.24399>
9 **CONF****JT-SAFE-V2: Safety-by-Design Foundation Model with World-Context Data**

ArXiv CS.AI 50%

arXiv:2605.24414v1 Announce Type: new Abstract: We introduce JT-Safe-V2, a large language model designed to advance the safety and trustworthiness of foundation models, extending our previous JT-Safe model toward a more comprehensive safety-by-design paradigm. JT-Safe-V2...

<https://arxiv.org/abs/2605.24414>
10 **CONF****Evolutionary Enhanced Multi-Agent Reinforcement Learning for Cooperative Air Combat**

ArXiv CS.AI 50%

arXiv:2605.25091v1 Announce Type: new Abstract: As modern air combat evolves toward beyond-visual-range (BVR) multi-aircraft cooperative engagements, autonomous decision-making for unmanned combat aerial vehicles (UCAVs) faces significant challenges due to high-dimensional state...

<https://arxiv.org/abs/2605.25091>
11 **CONF****Market Regime Council for Dynamic Credit Assignment in Multi-Agent LLM Decision Systems**

ArXiv CS.AI 50%

arXiv:2605.24490v1 Announce Type: new Abstract: Multi-agent LLM decision systems for portfolio management still lack a principled way to assign credit across specialist agents, remain vulnerable to cold-start dominance under regime shifts, and offer limited transparency into how...

<https://arxiv.org/abs/2605.24490>
12 **CONF****Reasoning as an Attack Surface: Adaptive Evolutionary CoT Jailbreaks for LLMs**

ArXiv CS.AI 50%

arXiv:2605.24497v1 Announce Type: new Abstract: Large Reasoning Models (LRMs) have demonstrated remarkable capabilities in reasoning and generation tasks and are increasingly deployed in real-world applications. However, their explicit chain-of-thought (CoT) mechanism introduces...

<https://arxiv.org/abs/2605.24497>

3 CONF

Emission-Aware Reinforcement Learning for Sustainable Electric Vehicle Charging and Carbon Dioxide Reduction Under Varying Renewable Penetration

ArXiv CS.AI 50%

arXiv:2605.24543v1 Announce Type: new Abstract: The rapid growth of Electric Vehicle (EV) adoption challenges power distribution networks through peak load spikes, voltage instability, and transformer overloads from uncoordinated charging. While Model Predictive Control (MPC)...

<https://arxiv.org/abs/2605.24543>

4 CONF

Jailbreak to Protect: Buffering and Reinforcing via Temporary Jailbreaking for Safe Fine-Tuning in Large Language Models

ArXiv CS.AI 50%

arXiv:2605.24550v1 Announce Type: new Abstract: Fine-tuning-as-a-Service (FaaS) enables personalization of large language models (LLMs), but it can weaken safety-alignment under harmful fine-tuning attacks. Recent work has shown that activating harmful-behavior modules during...

<https://arxiv.org/abs/2605.24550>

5 CONF

Associations between echocardiographic traits and AI-ECG predictions of heart failure

ArXiv CS.AI 50%

arXiv:2605.24576v1 Announce Type: new Abstract: Artificial intelligence-enabled electrocardiography (AI-ECG) can detect heart failure (HF), including disease not captured by left ventricular ejection fraction (LVEF), but the cardiac phenotypes underlying model predictions remain...

<https://arxiv.org/abs/2605.24576>

6 CONF

Remote sensing data imputation using deep learning for multispectral imagery

ArXiv CS.AI 50%

arXiv:2605.24003v1 Announce Type: cross Abstract: Remote sensing techniques have been increasingly utilised in aquatic applications in recent years. A common challenge in using optical satellite data is the presence of missing observations due to cloud cover. These data gaps can...

<https://arxiv.org/abs/2605.24003>

7 CONF

Learning to Reason Efficiently with A* Post-Training

ArXiv CS.AI 50%

arXiv:2605.24597v1 Announce Type: new Abstract: Many applications of large language models (LLMs) require deductive reasoning, yet models frequently produce incorrect or redundant inference steps. We frame natural language inference as a search problem where the final answer is...

<https://arxiv.org/abs/2605.24597>

8 CONF

LipoAgent: Coordinating Fine-Tuned LLM Agents for Safer Lipid Design

ArXiv CS.AI 50%

arXiv:2605.25250v1 Announce Type: new Abstract: Lipid nanoparticles (LNPs) are among the most clinically mature platforms for nucleic acid delivery, yet designing lipids that are both effective and biologically safe remains a major bottleneck. In practical screening, toxicity is...

<https://arxiv.org/abs/2605.25250>

- 9

CONF

Lattice theory and algebraic models for deep convolutional learning based on mathematical morphology

ArXiv CS.AI 50%

arXiv:2605.24608v1 Announce Type: new Abstract: We develop a rigorous algebraic framework for deep convolutional architectures, CNNs, ResNets, and encoder--decoder networks such as UNet, grounded in lattice theory and mathematical morphology. The central tool is the...

<https://arxiv.org/abs/2605.24608>
- 0

CONF

FloorplanQA: A Benchmark for Spatial Reasoning in LLMs using Structured Representations

ArXiv CS.AI 50%

arXiv:2507.07644v4 Announce Type: replace Abstract: We introduce FloorplanQA, a diagnostic benchmark for evaluating spatial reasoning in large language models (LLMs). FloorplanQA is grounded in structured representations of indoor scenes, such as (e.g., kitchens, living rooms,...

<https://arxiv.org/abs/2507.07644>
- 1

CONF

Beyond Inference-Only Deployment: Comparing Weight-Based Consolidation Against Cascading Compaction

ArXiv CS.AI 50%

arXiv:2605.24657v1 Announce Type: new Abstract: Major LLM platforms deploy models in an inference-only configuration: the model serves requests but never updates per-user weights. Users must repeatedly re-teach preferences, corrections, and project context, and context-based...

<https://arxiv.org/abs/2605.24657>
- 2

CONF

Measuring Reasoning Quality in LLMs: A Multi-Dimensional Behavioral Framework

ArXiv CS.AI 50%

arXiv:2605.24661v1 Announce Type: new Abstract: LLMs have achieved remarkable success in complex reasoning tasks, yet current evaluation approaches predominantly rely on final-answer correctness, offering limited insight into the underlying reasoning processes that produce those...

<https://arxiv.org/abs/2605.24661>
- 3

CONF

Credit Assignment with Resets in Language Model Reasoning

ArXiv CS.AI 50%

arXiv:2605.25507v1 Announce Type: new Abstract: Contemporary reinforcement learning with verifiable reward methods post-train language models on multi-step reasoning by assigning a single outcome reward uniformly across all tokens in a trajectory. Such uniform assignment ignores...

<https://arxiv.org/abs/2605.25507>
- 4

CONF

Emotional intelligence in large language models is fragmented across perception, cognition, and interaction

ArXiv CS.AI 50%

arXiv:2605.24686v1 Announce Type: new Abstract: As large language models (LLMs) are increasingly integrated into emotionally sensitive domains, the structural integrity of their emotional intelligence (EI) becomes a critical frontier for safety and alignment. Current benchmarks...

<https://arxiv.org/abs/2605.24686>

5 **CONF****Fundamental Limitation in Explaining AI**

ArXiv CS.AI 50%

arXiv:2605.24727v1 Announce Type: new Abstract: While large-scale models such as LLMs and diffusion models have achieved practical success, public institutions have emphasized the importance of explainability in AI. Existing methods for explaining AI, however, are not designed...

<https://arxiv.org/abs/2605.24727>

6 **CONF****Catching MRI outliers: unsupervised detection and localization of MRI artefacts and clinical anomalies using deep learning**

ArXiv CS.AI 50%

arXiv:2605.24609v1 Announce Type: cross Abstract: Artificial intelligence is increasingly integrated into radiotherapy workflows, yet such pipelines remain vulnerable to out-of-distribution image data that may introduce unexpected behavior in clinical tasks. Deep learning-based...

<https://arxiv.org/abs/2605.24609>

7 **CONF****Uncertainty Decomposition via Cyclical SG-MCMC and Soft-label Learning for Subjective NLP**

ArXiv CS.AI 50%

arXiv:2605.24773v1 Announce Type: new Abstract: Annotator disagreement in emotion classification reflects ambiguity intrinsic to emotion concepts and is essential for predictor-quality assessment in subjective NLP. Yet no prior work integrates soft-label learning with Bayesian...

<https://arxiv.org/abs/2605.24773>

8 **CONF****TriVAL: A Tri-Validation Framework for Faithful Automatic Optimization Modeling**

ArXiv CS.AI 50%

arXiv:2605.23966v1 Announce Type: cross Abstract: Optimization modeling serves as the pivotal bridge between natural-language problem descriptions and optimization solvers, and remains a cornerstone for bringing operations research (OR) into real-world decision making. Recent...

<https://arxiv.org/abs/2605.23966>

9 **CONF****PANDO: Efficient Multimodal AI Agents via Online Skill Distillation**

ArXiv CS.AI 50%

arXiv:2605.24785v1 Announce Type: new Abstract: Recent advances in multimodal web agents often rely on increased inference-time computation, including rollout search, verifier passes, offline skill discovery, and specialist model stacks. This raises a central question: can a web...

<https://arxiv.org/abs/2605.24785>

0 **CONF****Geo-Expert: Towards Expert-Level Geological Reasoning via Parameter-Efficient Fine-Tuning**

ArXiv CS.AI 50%

arXiv:2605.24844v1 Announce Type: new Abstract: While general-purpose Large Language Models (LLMs) applied to Geology often hallucinate when reasoning about subsurface structures and deep-time evolution, current AI in Earth sciences predominantly targets surface remote sensing...

<https://arxiv.org/abs/2605.24844>

1 **CONF****Energy Shields for Fairness**

ArXiv CS.AI 50%

arXiv:2605.24926v1 Announce Type: new Abstract: Runtime fairness is not a one-time constraint but a dynamic property evaluated over a sequence of decisions. To ensure fairness at runtime, it is necessary to account for past decisions, information neglected by conventional,...

<https://arxiv.org/abs/2605.24926>

2 **CONF****Cross-Domain Generalization Limits of Vision Foundation Models in Facial Deepfake Detection**

ArXiv CS.AI 50%

arXiv:2605.24965v1 Announce Type: cross Abstract: The rapid evolution of generative models has enabled the creation of hyper-realistic facial deepfakes, exposing a critical vulnerability in modern digital forensics: the inability of detectors to generalize to unseen manipulation...

<https://arxiv.org/abs/2605.24965>

3 **CONF****NeurIPS: Neuro-anatomical Inductive Priors for Sphere-based Brain Decoding**

ArXiv CS.AI 50%

arXiv:2605.24993v1 Announce Type: new Abstract: Current fMRI decoders face a performance-fidelity trade-off where efficient ID encoders outperform geometrically faithful surface-based models. We argue this is partly driven by inefficient surface tokenization and the failure to...

<https://arxiv.org/abs/2605.24993>

4 **CONF****Privacy-Preserving Local Language Models for Longitudinal Data Retrieval in Chronic Dermatologic Disease: Implementation in Pemphigus Patients**

ArXiv CS.AI 50%

arXiv:2605.25020v1 Announce Type: new Abstract: Chronic dermatologic diseases such as pemphigus require long-term follow-up, generating extensive longitudinal clinical documentation that is difficult to review comprehensively during routine visits and increasing clinician...

<https://arxiv.org/abs/2605.25020>

5 **CONF****Emergent Analogical Reasoning in Transformers**

ArXiv CS.AI 50%

arXiv:2602.01992v4 Announce Type: replace Abstract: Analogy is a central faculty of human intelligence, enabling abstract patterns discovered in one domain to be applied to another. Despite its central role in cognition, the mechanisms by which Transformers acquire and implement...

<https://arxiv.org/abs/2602.01992>

6 **CONF****Trust but Verify: Prover-Verifier Deliberation for Selective LLM Prediction**

ArXiv CS.AI 50%

arXiv:2605.25133v1 Announce Type: new Abstract: Reliably knowing when a language model is correct is almost as important as being correct. We introduce prover-verifier deliberation (PVD), an inference-time protocol grounded in interactive proof theory, as a mechanism for...

<https://arxiv.org/abs/2605.25133>

7 **CONF****From Latent Space to Training Data: Explainable Specialization in Minimal MLPs**

ArXiv CS.AI 50%

arXiv:2605.25939v1 Announce Type: cross Abstract: We here study whether training biases can make hidden neurons specialize in minimal one-hidden-layer MLPs, and whether such specialization improves prototype-based reconstruction of the training dataset from the learned weights....

<https://arxiv.org/abs/2605.25939>
8 **CONF****SimuWoB: Simulating Real-World Mobile Apps for Fast and Faithful GUI Agent Benchmarking**

ArXiv CS.AI 50%

arXiv:2605.25160v1 Announce Type: new Abstract: Mobile GUI agents powered by large language models have progressed rapidly, creating urgent needs for realistic and comprehensive evaluation. Existing benchmarks prioritize reproducibility but are often limited to open-source apps...

<https://arxiv.org/abs/2605.25160>
9 **CONF****DarkForest: Less Talk, Higher Accuracy for Multi-Agent LLMs**

ArXiv CS.AI 50%

arXiv:2605.25188v1 Announce Type: new Abstract: Multi-agent LLM systems improve reasoning by combining outputs from multiple agents, but interaction-heavy methods can introduce error propagation and high communication overhead. When agents exchange raw responses or reasoning...

<https://arxiv.org/abs/2605.25188>
10 **CONF****Boosting Inference with Guided Reasoning: Stochastic Exploration for Recursive Models**

ArXiv CS.AI 50%

arXiv:2605.25230v1 Announce Type: new Abstract: Recent work on recursive architectures has shown that tiny neural networks can be surprisingly powerful on structured reasoning tasks. The trick is to model reasoning trajectories with a latent dynamical system. We argue that the...

<https://arxiv.org/abs/2605.25230>
1 **CONF****FrontierOR: Benchmarking LLMs' Capacity for Efficient Algorithm Design in Large-Scale Optimization**

ArXiv CS.AI 50%

arXiv:2605.25246v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly used for optimization modeling and solver-code generation, yet practical operations research and optimization problems often require a harder capability: designing scalable algorithms...

<https://arxiv.org/abs/2605.25246>
2 **CONF****Whose Alignment? Comparing LLM Process Alignment Across Diverse Organizational Decision Contexts**

ArXiv CS.AI 50%

arXiv:2605.25256v1 Announce Type: new Abstract: Aligning AI systems with organizational decision-making is typically framed as a single-target problem: make the model behave like the organization. We argue this framing obscures a deeper pluralistic challenge. We rely on a...

<https://arxiv.org/abs/2605.25256>

3 CONF

AI Cartography: Mapping the Latent Landscape of AI Benchmark Ecosystems

ArXiv CS.AI 50%

arXiv:2605.25272v1 Announce Type: new Abstract: While aggregate leaderboard scores drive AI development, they contain substantial measurement noise whose sources and magnitudes remain unquantified, making it unclear when rankings reflect genuine capability differences versus...

<https://arxiv.org/abs/2605.25272>

4 CONF

Towards end-to-end LLM-based censoring-aware survival analysis

ArXiv CS.AI 50%

arXiv:2605.25399v1 Announce Type: new Abstract: Objective: Survival analysis is central to medical prediction, yet large language models (LLMs) are rarely used as end-to-end survival models because censoring prevents straightforward supervised fine-tuning. Here we present...

<https://arxiv.org/abs/2605.25399>

5 CONF

A Signal-Language Foundation Model for Broad-Spectrum Cardiovascular Assessment from Routine Electrocardiography

ArXiv CS.AI 50%

arXiv:2605.25446v1 Announce Type: new Abstract: Electrocardiography (ECG) is central to cardiovascular care, but conventional AI models are often restricted to common arrhythmias and may generalize poorly across populations or clinically subtle diseases. We developed ECG...

<https://arxiv.org/abs/2605.25446>

6 CONF

What Gets Cited: Competitive GEO in AI Answer Engines

ArXiv CS.AI 50%

arXiv:2605.25517v1 Announce Type: new Abstract: AI answer engines generate answers from retrieved pages but cite only a few sources. This makes visibility depend not just on ranking, but on being cited. We study competitive Generative Engine Optimization (GEO): when two...

<https://arxiv.org/abs/2605.25517>

7 CONF

StructBreak: Structural Cognitive Overload-Induced Safety Failures in MLLMs

ArXiv CS.AI 50%

arXiv:2605.25534v1 Announce Type: new Abstract: Multimodal Large Language Models (MLLMs) excel at structural reasoning yet suffer from a sharp logical brittleness in structural consistency. We term this phenomenon Structural Cognitive Overload (SCO), a byproduct of the...

<https://arxiv.org/abs/2605.25534>

8 CONF

ADMFormer: An Adaptive-Decomposition Transformer with Time-Varying Masked Spatial Attention for Traffic Forecasting

ArXiv CS.AI 50%

arXiv:2605.25543v1 Announce Type: new Abstract: Accurate traffic forecasting is essential for intelligent transportation systems, supporting a wide range of real-world applications. However, it remains challenging due to two key factors:~(1) Traffic series contain heterogeneous...

<https://arxiv.org/abs/2605.25543>

9 CONF

Uncertainty Reasoning with Large Language Models for Explainable Disease Diagnosis

ArXiv CS.AI 50%

arXiv:2605.25566v1 Announce Type: new Abstract: Clinical decision-making requires reasoning over incomplete, imprecise, and linguistically expressed patient narratives. While large language models (LLMs) excel at extracting latent information from natural language, they lack the...

<https://arxiv.org/abs/2605.25566>

0 CONF

Detecting Unfaithful Chain-of-Thought via Circuit-Guided Internal-External Discrepancy

ArXiv CS.AI 50%

arXiv:2605.25603v1 Announce Type: new Abstract: Chain-of-thought (CoT) reasoning improves the problem-solving ability of large language models (LLMs), but generated reasoning traces may not faithfully reflect the model's actual decision process. Existing CoT unfaithfulness...

<https://arxiv.org/abs/2605.25603>

1 CONF

CausaLab: A Scalable Environment for Interactive Causal Discovery Toward AI Scientists

ArXiv CS.AI 50%

arXiv:2605.26029v1 Announce Type: new Abstract: We introduce CausaLab, a scalable environment for evaluating interactive causal discovery by LLM agents. Unlike prior evaluations, CausaLab evaluates both whether an agent can solve a problem using causal evidence and whether its...

<https://arxiv.org/abs/2605.26029>

2 CONF

AgentHijack: Benchmarking Computer Use Agent Robustness to Common Environment Corruptions

ArXiv CS.AI 50%

arXiv:2605.25707v1 Announce Type: new Abstract: Autonomous computer use agents that powered by multimodal large language models (MLLMs) are emerging as capable assistants for completing complex digital workflows. However, real-world execution environments are far from ideal:...

<https://arxiv.org/abs/2605.25707>

3 CONF

MuCRASP: Multimodal Chain-of-thought Reasoning aware Structured Pruning

ArXiv CS.AI 50%

arXiv:2605.25842v1 Announce Type: new Abstract: Vision-language models (VLMs) increasingly rely on chain-of-thought (CoT) reasoning to solve complex multimodal tasks, but their large parameter sizes make deployment expensive. Structured pruning offers a natural solution;...

<https://arxiv.org/abs/2605.25842>

4 CONF

\$D^2\$-Monitor: Dynamic Safety Monitoring for Diffusion LLMs via Hesitation-Aware Routing

ArXiv CS.AI 50%

arXiv:2605.25893v1 Announce Type: new Abstract: Despite the emergence of diffusion large language models (D-LLMs) as an alternative to autoregressive large language models (AR-LLMs), safety monitoring for D-LLMs remains largely unexplored. Unlike AR-LLMs, D-LLMs generate text...

<https://arxiv.org/abs/2605.25893>

5 **CONF****CITYREP: A Unified Benchmark for Urban Representations Across Cities, Tasks, and Modalities****ArXiv CS.AI 50%**

arXiv:2605.26036v1 Announce Type: new Abstract: Urban representation learning encodes complex urban environments into general-purpose embeddings for diverse downstream tasks and emerging urban foundation models. However, current evaluations are limited, typically focusing on one...

<https://arxiv.org/abs/2605.26036>

6 **CONF****Local MAP Sampling for Diffusion Models****ArXiv CS.AI 50%**

arXiv:2510.07343v3 Announce Type: replace-cross Abstract: Diffusion Posterior Sampling (DPS) provides a principled Bayesian approach to inverse problems by sampling from $p(x_0 \mid y)$. While posterior sampling is valuable for capturing uncertainty and multi-modality, many...

<https://arxiv.org/abs/2510.07343>

7 **CONF****Claw-Anything: Benchmarking Always-On Personal Assistants with Broader Access to User's Digital World****ArXiv CS.AI 50%**

arXiv:2605.26086v1 Announce Type: new Abstract: Large language model agents are increasingly envisioned as always-on personal assistants with access to anything relevant in the user's digital world. Yet current systems operate over only narrow slices of that world, limiting...

<https://arxiv.org/abs/2605.26086>

8 **CONF****From Model Scaling to System Scaling: Scaling the Harness in Agentic AI****ArXiv CS.AI 50%**

arXiv:2605.26112v1 Announce Type: new Abstract: This paper studies the next major bottleneck in agentic AI as system scaling, not only model scaling: the design of auditable, persistent, modular, and verifiable architectures around foundation models. We refer to this shift as...

<https://arxiv.org/abs/2605.26112>

9 **CONF****Tokenizer Fertility and Zero-Shot Performance of Foundation Models on Ukrainian Legal Text: A Comparative Study****ArXiv CS.AI 50%**

arXiv:2605.14890v2 Announce Type: cross Abstract: Tokenizer fertility varies 1.6x across foundation models on Ukrainian legal text, yet this cost-critical dimension is absent from model selection practice. We benchmark seven models from five providers on 273 validated court...

<https://arxiv.org/abs/2605.14890>

0 **CONF****Check Your LLM's Secret Dictionary! Five Lines of Code Reveal What Your LLM Learned (Including What It Shouldn't Have)****ArXiv CS.AI 50%**

arXiv:2605.22005v1 Announce Type: cross Abstract: We show that singular value decomposition of the `lm_head` weight matrix of a transformer-based large language model -- requiring only five lines of PyTorch and no model inference -- reveals interpretable semantic subspaces...

<https://arxiv.org/abs/2605.22005>

1 **CONF****AI-Driven Alpha Decay: Algorithmic Homogenization, Reflexive Signal Erosion, and the Paradox of Intelligent Markets**

ArXiv CS.AI 50%

arXiv:2605.23905v1 Announce Type: cross Abstract: We show that AI-driven investment strategies are inherently self-defeating at scale. As AI adoption rises, three mutually reinforcing channels -- signal crowding, performative signal erosion, and Red Queen competition -- compress...

<https://arxiv.org/abs/2605.23905>
2 **CONF****Document Classification Pattern Recognition via Information Fusion: A Systematic Review of Multimodal and Multiview Representation Approaches**

ArXiv CS.AI 50%

arXiv:2605.23910v1 Announce Type: cross Abstract: Information fusion is used widely to improve document classification by the integration of multiple data sources (multimodal) or representations (multiview). However, the field lacks a unified framework, a quantitative synthesis...

<https://arxiv.org/abs/2605.23910>
3 **CONF****FG-CLIP 2: A Bilingual Fine-grained Vision-Language Alignment Model**

ArXiv CS.AI 50%

arXiv:2510.10921v3 Announce Type: replace-cross Abstract: Fine-grained vision-language understanding requires precise alignment between visual content and linguistic descriptions, a capability that remains limited in current models, particularly in non-English settings. While...

<https://arxiv.org/abs/2510.10921>
4 **CONF****Agent-Facing Information Design in LLM Tool Registries**

ArXiv CS.AI 50%

arXiv:2605.23916v1 Announce Type: cross Abstract: LLM tool registries function as unregulated advertising platforms: providers write free-text descriptions that agents use for selection, yet no measurement infrastructure -- no viewability standard, quality score, or outcome...

<https://arxiv.org/abs/2605.23916>
5 **CONF****Authority Signals in Claude AI Health Citations: A Descriptive Analysis Using the Authority Signals Framework**

ArXiv CS.AI 50%

arXiv:2605.23921v1 Announce Type: cross Abstract: This study seeks to determine the authority signals used by Anthropic's Claude AI in its presentation of sources when answering consumer health questions. While there exists a great deal of discourse around the quality of health...

<https://arxiv.org/abs/2605.23921>
6 **CONF****From Multi-Agent Systems and the Semantic Web to Agentic AI: A Unified Narrative of the Web of Agents**

ArXiv CS.AI 50%

arXiv:2507.10644v4 Announce Type: replace Abstract: The Web of Agents (WoA) transforms the document-centric Web into an environment of autonomous agents acting on users' behalf, a vision newly tractable as large language models (LLMs) mature. We argue that across three decades...

<https://arxiv.org/abs/2507.10644>

7 CONF

Catching The Correct Answer Trap: Characterising AI Tutor Blind Spots When Analysing Student Reasoning

ArXiv CS.AI 50%

arXiv:2605.23925v1 Announce Type: cross Abstract: Intelligent tutoring systems increasingly provide automated feedback on student work, but robust feedback requires assessing reasoning, not only final answers. We study a failure mode we call the correct answer trap (CAT): models...

<https://arxiv.org/abs/2605.23925>

8 CONF

AI-Driven Controlled Environment Agriculture as Resilient Infrastructure for U.S. Fresh-Produce Supply Chains

ArXiv CS.AI 50%

arXiv:2605.23946v1 Announce Type: cross Abstract: Climate volatility, regional production concentration, labor constraints, cyber risk, and dependence on long-distance fresh-produce supply chains expose vulnerabilities in U.S. fresh-produce and specialty-crop systems. Controlled...

<https://arxiv.org/abs/2605.23946>

9 CONF

OASIS: Observation-Action Space Alignment via SE(3) Trajectory Prediction for Robotic Manipulation

ArXiv CS.AI 50%

arXiv:2605.25829v1 Announce Type: cross Abstract: Recent vision-language-action (VLA) models and world action models (WAMs) advance robotic manipulation by enriching intermediate representations with auxiliary spatial features or future visual-state prediction. However, these...

<https://arxiv.org/abs/2605.25829>

00 CONF

AI in the Enterprise: How People Use M365 Copilot Chat

ArXiv CS.AI 50%

arXiv:2605.23958v1 Announce Type: cross Abstract: M365 Copilot is used every week by millions of people across more than a million companies around the world as part of their workflows. Uniquely positioned in the AI landscape given its near-exclusive use for work purposes, M365...

<https://arxiv.org/abs/2605.23958>

01 CONF

Multimodal Alignment and Preference Optimization for Zero-Shot Conditional RNA Generation

ArXiv CS.AI 50%

arXiv:2605.23961v1 Announce Type: cross Abstract: The design of RNA molecules that interact with specific proteins is a critical challenge in experimental and computational biology. Despite recent progress in natural language modeling and deep learning-based protein design,...

<https://arxiv.org/abs/2605.23961>

02 CONF

Sensing Intelligence as a Trainable Metamaterial Property

ArXiv CS.AI 50%

arXiv:2605.23967v1 Announce Type: cross Abstract: In biological systems, sensing is not performed by the brain alone: the body deforms, vibrates, and filters external stimuli before they are transduced into neural signals. In engineered systems, this processing burden is placed...

<https://arxiv.org/abs/2605.23967>

03 CONF

Parameter Efficient Multi-Class Intelligent Scheduling for Multimodal Online Distributed Industrial Anomaly Detection

ArXiv CS.AI 50%

arXiv:2605.23984v1 Announce Type: cross Abstract: Industrial anomaly detection has attracted significant attention as a fundamental challenge in industrial systems. The rapid advancement of heterogeneous industrial sensors has driven industrial anomaly detection from unimodal to...

<https://arxiv.org/abs/2605.23984>

04 CONF

RAW: Robust Avatar Watermarking -- Benchmarking and Baseline

ArXiv CS.AI 50%

arXiv:2605.23994v1 Announce Type: cross Abstract: Digital avatar watermarking presents unique challenges: avatars are routinely post-processed with background replacement, reframing, and format conversion before deployment. We introduce \textbf{RAW} (Robust Avatar Watermarking),...

<https://arxiv.org/abs/2605.23994>

05 CONF

SA-Kura: An Energy-Efficient Systolic Array Accelerator for Locally-Coupled Kuramoto Drift in Diffusion Sampling

ArXiv CS.AI 50%

arXiv:2605.24016v1 Announce Type: cross Abstract: Diffusion inference remains costly for edge deployment, yet existing accelerators focus almost exclusively on score networks because standard drift is merely a trivial linear scaling. Kuramoto orientation diffusion replaces this...

<https://arxiv.org/abs/2605.24016>

06 CONF

WTKO-CNN: Deep Learning Reveals Sequence Motifs Distinguishing Wild-Type and Knockout ATAC-seq Peaks

ArXiv CS.AI 50%

arXiv:2605.24034v1 Announce Type: cross Abstract: Chromatin regulators can alter transcriptional programs by modifying the accessibility of regulatory DNA elements. Understanding how regulatory sequences differ between wild-type (WT) and knockout (KO) conditions is crucial for...

<https://arxiv.org/abs/2605.24034>

07 CONF

Mode-as-Sequence: Translating Multimodal Motion Prediction into Unified Sequential Mode Modeling

ArXiv CS.AI 50%

arXiv:2605.24037v1 Announce Type: cross Abstract: Multimodal motion forecasting is inherently under-supervised: each training scene provides only one realized future, yet multiple plausible futures exist. This sparse supervision often leads to mode collapse (redundant hypotheses...

<https://arxiv.org/abs/2605.24037>

08 CONF

LLM-AutoSciLab: Closed-Loop Scientific Discovery via Active Experimentation with LLMs

ArXiv CS.AI 50%

arXiv:2605.24043v1 Announce Type: cross Abstract: Scientific discovery is a closed-loop process in which hypotheses guide data acquisition and observations refine the hypothesis space. Yet most approaches reduce discovery to supervised learning over fixed datasets, where limited...

<https://arxiv.org/abs/2605.24043>

09 CONF

A Large-Scale Dataset and Benchmark: Do Protein-Ligand Models Learn Binding Sites or Just Binding Likelihood?

ArXiv CS.AI 50%

arXiv:2605.24045v1 Announce Type: cross Abstract: Protein-ligand modeling underpins computational drug discovery and molecular design. Existing protein-ligand benchmarks typically evaluate whether a protein and ligand interact and how strongly they bind, through tasks such as...

<https://arxiv.org/abs/2605.24045>

10 CONF

Mixture of Complementary Agents for Robust LLM Ensemble

ArXiv CS.AI 50%

arXiv:2605.24048v1 Announce Type: cross Abstract: Multi-AI collaboration, such as ensembling or debating large language models (LLMs), is a promising paradigm for aggregating information and boosting performance. A foundational step in these pipelines is to feed the responses of...

<https://arxiv.org/abs/2605.24048>

11 CONF

Truthful Online Preference Aggregation for LLM Fine-Tuning in Mobile Crowdsourcing

ArXiv CS.AI 50%

arXiv:2605.24052v1 Announce Type: cross Abstract: To better serve users' demands in mobile applications (e.g., navigation), mobile crowdsourcing platforms can iteratively align large language model (LLM)-generated content (e.g., AI-generated traffic condition predictions) with...

<https://arxiv.org/abs/2605.24052>

12 CONF

Grammatically-Guided Sparse Attention for Efficient and Interpretable Transformers

ArXiv CS.AI 50%

arXiv:2605.24518v1 Announce Type: cross Abstract: The quadratic complexity of self-attention in Transformer models remains a significant bottleneck for processing long sequences and deploying large language models efficiently. For this approach, there has been significant...

<https://arxiv.org/abs/2605.24518>

13 CONF

Generative Representation Learning on Hyper-relational Knowledge Graphs via Masked Discrete Diffusion

ArXiv CS.AI 50%

arXiv:2605.24064v1 Announce Type: cross Abstract: Hyper-relational knowledge graphs (HKGs) effectively represent complex facts. While inferring new knowledge in HKGs is a critical problem, current methods cast it as a simple link prediction, assuming that nearly all entities and...

<https://arxiv.org/abs/2605.24064>

14 CONF

When the Manual Lies: A Realistic Benchmark to Evaluate MCP Poisoning Attacks for LLM Agents

ArXiv CS.AI 50%

arXiv:2605.24069v1 Announce Type: cross Abstract: The rise of tool-using Large Language Model (LLM) agents, standardized by protocols like the Model Context Protocol (MCP), has unlocked unprecedented autonomous execution capabilities for LLM Agents by integrating external...

<https://arxiv.org/abs/2605.24069>

15 CONF

Verified SHAP: Provable Bounds for Exact Shapley Values of Neural Networks

ArXiv CS.AI 50%

arXiv:2605.24084v1 Announce Type: cross Abstract: Shapley additive explanations (SHAP) are widely recognised as computationally intractable for neural networks, since they induce an exponential search space over the input features. In this work, we take a first step towards...

<https://arxiv.org/abs/2605.24084>

16 CONF

Overcoming "Physics Shock" in Earth Observation A Heteroscedastic Uncertainty Framework for PINN-based Flood Inference

ArXiv CS.AI 50%

arXiv:2605.24106v1 Announce Type: cross Abstract: Rapid and accurate flood extent mapping from Remote Sensing data, such as Synthetic Aperture Radar (SAR), is critical for operational disaster response, but standard Deep Learning models often produce physically impossible...

<https://arxiv.org/abs/2605.24106>

17 CONF

An Empirical Evaluation of LLM-Generated Code Security Across Prompting Methods

ArXiv CS.AI 50%

arXiv:2605.24298v1 Announce Type: cross Abstract: The growing use of Large Language Models (LLMs) for automated code generation has enhanced software development efficiency, but often at the cost of security. Generated code frequently overlooks critical concerns, leaving it...

<https://arxiv.org/abs/2605.24298>

18 CONF

Knowledge Graph Modulated Deep Learning for Limited-Sample Clinical Data Analysis

ArXiv CS.AI 50%

arXiv:2605.24162v1 Announce Type: cross Abstract: Biological systems are governed by structured molecular interactions, where pathways, regulatory circuits, and functional gene relationships shape cellular behavior and disease progression. Much of this knowledge is naturally...

<https://arxiv.org/abs/2605.24162>

19 CONF

PromptAudit: Auditing Prompt Sensitivity in LLM-Based Vulnerability Detection

ArXiv CS.AI 50%

arXiv:2605.24171v1 Announce Type: cross Abstract: Large language models are increasingly used for vulnerability detection, yet their reliability under different prompt formulations remains uncharacterized. We present PromptAudit, a controlled evaluation framework that isolates...

<https://arxiv.org/abs/2605.24171>

20 CONF

Filtered Posterior Mean Collections: A Unified Framework for Analytical Models of Diffusion Generalization

ArXiv CS.AI 50%

arXiv:2605.24192v1 Announce Type: cross Abstract: The neural-network denoising functions which form the backbone of image diffusion models are remarkably consistent in their generalization behaviour across a wide variety of network architectures and training procedure...

<https://arxiv.org/abs/2605.24192>

21 CONF

Improving Labeling Consistency with Detailed Constitutional Definitions and AI-Driven Evaluation

ArXiv CS.AI 50%

arXiv:2605.24247v1 Announce Type: cross Abstract: Many automated labeling pipelines classify inputs into categories defined by a written specification, content moderation being a prominent use case. Simple category definitions are not detailed enough for labelers to produce the...

<https://arxiv.org/abs/2605.24247>

22 CONF

Benchmarking Patent Embeddings: A Multi-Task Evaluation of 22 Models Across Retrieval, Classification, and Clustering

ArXiv CS.AI 50%

arXiv:2605.24297v1 Announce Type: cross Abstract: Which fine-tuning signals improve patent embedding models, and do gains transfer across patent landscapes? We benchmark 22 embedding models, from 22M-parameter encoders to 12B instruction-tuned LLMs, on retrieval, classification,...

<https://arxiv.org/abs/2605.24297>

23 CONF

ScaleAcross Explorer: Exploring Communication Optimization for Scale-Across AI Model Training

ArXiv CS.AI 50%

arXiv:2605.24326v1 Announce Type: cross Abstract: The rapid scaling of large language model training requires distributing GPU resources across multiple data center buildings and regions. We refer to such paradigm as "scale-across" training. As infrastructure expands, the system...

<https://arxiv.org/abs/2605.24326>

24 CONF

Assessing the Operational Viability of Foundation Models for Time Series Forecasting

ArXiv CS.AI 50%

arXiv:2605.24381v1 Announce Type: cross Abstract: Time series forecasting drives operational decisions in areas like finance, transportation, and energy. While supervised learning approaches achieve strong performance, they require domain-specific training, feature engineering,...

<https://arxiv.org/abs/2605.24381>

25 CONF

MX-SAFE: Versatile Inference- and Training-Proof Microscaling Format with On-the-Fly Exponent and Mantissa Bit Allocation

ArXiv CS.AI 50%

arXiv:2605.24391v1 Announce Type: cross Abstract: As the demand for deep learning grows, cost reduction through quantization has become essential for both training and inference. In 2022, the Open Compute Project (OCP) consortium standardized narrow precision formats for deep...

<https://arxiv.org/abs/2605.24391>

26 CONF

VectorArk: Learning Practical Image Vectorization with Rounded Polygon Representation

ArXiv CS.AI 50%

arXiv:2605.24398v1 Announce Type: cross Abstract: Recent vision-language model (VLM)-based approaches have achieved impressive results on image vectorization tasks. However, they are typically evaluated on synthetic benchmarks, where clean SVGs are rasterized at high resolution...

<https://arxiv.org/abs/2605.24398>

27 CONF

Temporal Concept Drift in Legal Judgment Prediction: Neural Baselines Across Three Epochs of Ukrainian Court Decisions

ArXiv CS.AI 50%

arXiv:2605.24452v1 Announce Type: cross Abstract: Legal NLP benchmarks evaluate models on randomly split data, implicitly assuming that legal language is stationary. We test this assumption by fine-tuning four transformer encoders -- XLM-RoBERTa (base and large) and their...

<https://arxiv.org/abs/2605.24452>

28 CONF

Code2UML: Agentic LLMs with context engineering for scalable software visualization

ArXiv CS.AI 50%

arXiv:2605.24453v1 Announce Type: cross Abstract: Large Language Model (LLM)-based code analysis tools are adopted to automate software documentation tasks. However, the scalability of these approaches to real codebases, where Intermediate Representations (IR) exceed LLM context...

<https://arxiv.org/abs/2605.24453>

29 CONF

Balancing Fairness, Privacy, and Accuracy: A Multitask Adversarial Framework for Centralized Data-Driven Systems

ArXiv CS.AI 50%

arXiv:2605.24458v1 Announce Type: cross Abstract: The integration of fairness and privacy in centralized data-driven applications is critical, especially as these systems increasingly influence sectors with significant societal impact. Current methods rarely address privacy,...

<https://arxiv.org/abs/2605.24458>

30 CONF

Coarse-to-Fine Domain Incremental Learning with Attentive Distillation for Mining Footprint Segmentation in Multispectral Imagery

ArXiv CS.AI 50%

arXiv:2605.24460v1 Announce Type: cross Abstract: Automatically mapping and segmenting global mining footprints using remote sensing and deep learning is critical for monitoring the socio-environmental risks and impacts of mining, yet its progress is hindered by the scarcity of...

<https://arxiv.org/abs/2605.24460>

31 CONF

{\Phi}-Noise: Training-Free Temporal Video Conditioning via Phase-Based Noise Manipulation

ArXiv CS.AI 50%

arXiv:2605.24509v1 Announce Type: cross Abstract: Latent video diffusion models generate videos by progressively transforming Gaussian noise into realistic samples conditioned on text or visual inputs. However, existing conditioning methods often require additional training and...

<https://arxiv.org/abs/2605.24509>

32 CONF

Is Decentralized AI Governable? From Regulatory Policy to Constitutive Protocol

ArXiv CS.AI 50%

arXiv:2605.24538v1 Announce Type: cross Abstract: Every major framework for governing artificial intelligence presupposes an identifiable entity -- a developer, deployer, or operator -- who can be held responsible and compelled to comply. Decentralized AI (DeAI) dissolves this...

<https://arxiv.org/abs/2605.24538>

33 CONF

SemanticZip: A Pilot Framework for Lossy Text Compression with LLMs as Semantic Decompressors

ArXiv CS.AI 50%

arXiv:2605.24541v1 Announce Type: cross Abstract: Text compression for large language model (LLM) systems is usually framed as token deletion, retrieval, summarization, or exact reconstruction. We study a more aggressive but explicitly lossy setting: compress text into compact...

<https://arxiv.org/abs/2605.24541>

34 CONF

AI-Driven Adaptive Adversaries and the Erosion of Cryptographic Trust in Public Key Systems

ArXiv CS.AI 50%

arXiv:2605.24542v1 Announce Type: cross Abstract: This paper examines the erosion of Public Key Cryptography (PKC) security under adaptive adversarial optimisation driven by artificial intelligence. The problem addressed is the growing mismatch between algorithm-centric...

<https://arxiv.org/abs/2605.24542>

35 CONF

Polymorphism Is Rotation: Operational Mechanistic Interpretability from a Two-Layer Transformer to Pythia-70m

ArXiv CS.AI 50%

arXiv:2605.24577v1 Announce Type: cross Abstract: Independently trained transformers compute the same function in residual-stream bases that differ by a uniform random rotation on $\mathrm{SO}(d_{\mathrm{model}})$. We call this phenomenon polymorphism: same function, mutually...

<https://arxiv.org/abs/2605.24577>

36 CONF

Guarded Repair for Harm-Aware Post-hoc Replacement of LLM Mathematical Reasoning

ArXiv CS.AI 50%

arXiv:2605.24613v1 Announce Type: cross Abstract: Post-hoc repair of LLM mathematical reasoning introduces an asymmetric risk: fixing an incorrect reasoning trace is useful, but replacing a trace that was already correct can be harmful. We study this problem under a selective...

<https://arxiv.org/abs/2605.24613>

37 CONF

Beyond Generative Priors: Minority Sampling with JEPA-Guided Diffusion

ArXiv CS.AI 50%

arXiv:2605.24631v1 Announce Type: cross Abstract: Minority sampling aims to generate low-density instances on a data manifold and is of central importance in applications such as medical diagnosis, anomaly detection, and creative AI. Existing approaches, however, define minority...

<https://arxiv.org/abs/2605.24631>

38 CONF

DisDop: Distillation with Domain Priors for Open-Vocabulary Aerial Object Detection

ArXiv CS.AI 50%

arXiv:2605.24639v1 Announce Type: cross Abstract: With the widespread application of drones in recent years, object detection of aerial images has attracted increasing attention, especially open-vocabulary aerial detection which is not restricted to predefined categories. Due to...

<https://arxiv.org/abs/2605.24639>

39 CONF

How Many Tools Should an LLM Agent See? A Chance-Corrected Answer

ArXiv CS.AI 50%

arXiv:2605.24660v1 Announce Type: cross Abstract: Before an LLM agent can use a tool, a retrieval system must decide which candidate tools to show to the agent. How long should that shortlist be? Show too many tools and the model struggles to choose. Show too few and the correct...

<https://arxiv.org/abs/2605.24660>

40 CONF

HoloFair: Unified T2I Fairness Evaluation and Fair-GRPO Debiasing

ArXiv CS.AI 50%

arXiv:2605.24687v1 Announce Type: cross Abstract: Text-to-Image (T2I) models have made significant strides in visual realism and semantic consistency, yet they often perpetuate and amplify societal biases. Existing evaluation methods typically address only single-dimensional...

<https://arxiv.org/abs/2605.24687>

41 CONF

The Path Matters: Learning a Token-Commitment Policy for Diffusion Language Models

ArXiv CS.AI 50%

arXiv:2605.24697v1 Announce Type: cross Abstract: Diffusion large language models promise faster generation by refining many token positions in parallel, but this parallelism introduces a hidden control problem: which proposed tokens should be transferred into the partially...

<https://arxiv.org/abs/2605.24697>

42 CONF

BackWeak: Backdoor Knowledge Distillation Simply with Weak Triggers and Fine-tuning

ArXiv CS.AI 50%

arXiv:2511.12046v2 Announce Type: replace-cross Abstract: Knowledge Distillation (KD) is essential for compressing large models, yet relying on pre-trained "teacher" models downloaded from third-party repositories introduces serious security risks--most notably backdoor attacks....

<https://arxiv.org/abs/2511.12046>

43 CONF

Spectral Retrieval: Multi-Scale Sinc Convolution over Token Embeddings for Localized Retrieval in LLM Multi-Agent Systems

ArXiv CS.AI 50%

arXiv:2605.24764v1 Announce Type: cross Abstract: [Abridged] - Spectral Retrieval is a plug-in re-ranking stage that interpolates between per-token MaxSim and mean-pool retrieval through a multi-scale sinc convolution over token embeddings. In standard dense retrieval each...

<https://arxiv.org/abs/2605.24764>

44 CONF

From Theory to Decision Rule: Calibrating the Noisy-Label Crossover for Vision-Language Model Weak Supervision Across Three Medical-Imaging Benchmarks

ArXiv CS.AI 50%

arXiv:2605.24771v1 Announce Type: cross Abstract: Classical noisy-label theory predicts that downstream performance under weak supervision is bounded above by the labeler's accuracy, implying a sharp crossover: once a gold-trained classifier matches the labeler, weak labels stop...

<https://arxiv.org/abs/2605.24771>

45 CONF

IndexMem: Learned KV-Cache Eviction with Latent Memory for Long-Context LLM Inference

ArXiv CS.AI 50%

arXiv:2605.25475v1 Announce Type: cross Abstract: Large Language Models (LLMs) are increasingly expected to operate over long contexts, yet standard softmax attention incurs a KV cache that grows linearly with sequence length, quickly becoming the bottleneck for long context...

<https://arxiv.org/abs/2605.25475>

46 CONF

Reflect-Guard: Enhancing LLM Safeguards against Adversarial Prompts via Logical Self-Reflection

ArXiv CS.AI 50%

arXiv:2605.24834v1 Announce Type: cross Abstract: Large language model (LLM) safety classifiers such as Llama Guard are effective at detecting overtly harmful prompts but remain vulnerable to adversarial jailbreak attacks that disguise malicious intent through role-play...

<https://arxiv.org/abs/2605.24834>

47 CONF

Tiny Brains, Giant Impact: Uncovering the Keystone Neurons of LLM with Just a Few Prompts

ArXiv CS.AI 50%

arXiv:2605.24846v1 Announce Type: cross Abstract: Large language models (LLMs) display strong comprehensive abilities, yet the internal mechanisms that support these behaviors remain insufficiently understood. In this work, we show that across a wide range of open-weight...

<https://arxiv.org/abs/2605.24846>

48 CONF

The Concept Allocation Zone: Tracking How Concepts Form Across Transformer Depth

ArXiv CS.AI 50%

arXiv:2605.24856v1 Announce Type: cross Abstract: Concept formation in transformer language models is depth-extended, not a single-layer event: concepts emerge gradually across a contiguous region of the residual stream. Mechanistic interpretability methods identify the single...

<https://arxiv.org/abs/2605.24856>

49 CONF

DBPnet: Damper Characteristics-Based Bayesian Physics-Informed Neural Network for Wheel Load Estimation

ArXiv CS.AI 50%

arXiv:2605.24860v1 Announce Type: cross Abstract: Advanced driver assistance systems (ADAS) play an important role in modern automotive intelligence, significantly enhancing vehicle safety and stability. The performance of ADAS critically relies on accurate and reliable vehicle...

<https://arxiv.org/abs/2605.24860>

50 CONF

On the Impact of Class Imbalance on the Learning Dynamics of Deep Neural Networks:An Intuitive Insight

ArXiv CS.AI 50%

arXiv:2605.24908v1 Announce Type: cross Abstract: Class imbalance in deep neural networks (DNNs) has witnessed a rapid increase in research attention in recent years. However, the varying accounts of the reasons behind the poor performance of DNN on imbalance data is pertinent...

<https://arxiv.org/abs/2605.24908>

51 CONF

Explainable Multi-Task Retinal Imaging Reveals Microvascular Signals for Systemic Risk Stratification in Type 2 Diabetes: A Pilot Study

ArXiv CS.AI 50%

arXiv:2605.24913v1 Announce Type: cross Abstract: Retinal imaging provides a non-invasive window into systemic microvascular health and has emerged as a potential biomarker for systemic diseases. However, whether retinal features encode biologically meaningful systemic signals...

<https://arxiv.org/abs/2605.24913>

52 CONF

Your Embedding Model is SMARTer Than You Think

ArXiv CS.AI 50%

arXiv:2605.24938v1 Announce Type: cross Abstract: Multimodal retrieval relies heavily on single-vector retrievers, which compress rich, sequential token sequences into one single global representation. While efficient, they discard fine-grained, local evidence critical for dense...

<https://arxiv.org/abs/2605.24938>

53 CONF

Riemannian-Manifold Steering: Geometry-Aware Generative Autoencoders for Label-Free Steering

ArXiv CS.AI 50%

arXiv:2605.24942v1 Announce Type: cross Abstract: Steering a language model - intervening on its internal activations to change downstream behaviour - has recently expanded beyond linear interpolation to nonlinear methods such as angular and kernelized steering, which define...

<https://arxiv.org/abs/2605.24942>

54 CONF

Efficient Benchmarking Is Just Feature Selection and Multiple Regression

ArXiv CS.AI 50%

arXiv:2605.25773v1 Announce Type: cross Abstract: Efficient benchmarking techniques aim to lower the computational cost of evaluating LLMs by predicting full benchmark scores using only a subset of a benchmark's questions. By reframing this problem as an instance of multiple...

<https://arxiv.org/abs/2605.25773>

55 CONF

Investigating the Interplay between Contextual and Parametric Chain-of-Thought Faithfulness under Optimization

ArXiv CS.AI 50%

arXiv:2605.24960v1 Announce Type: cross Abstract: Chain-of-Thought (CoT) faithfulness, i.e., whether CoTs genuinely reflect large language models' (LLM) underlying behavior, is typically evaluated under two disjoint paradigms: contextual faithfulness, measured by perturbing the...

<https://arxiv.org/abs/2605.24960>

56 CONF

Selective Test-Time Compute Scaling for Click-Through Rate Prediction via Uncertainty-Triggered Feature Path Exploration

ArXiv CS.AI 50%

arXiv:2605.24989v1 Announce Type: cross Abstract: Scaling test-time compute has proven highly effective for language models, yet this opportunity remains largely unexplored for industrial Click-Through Rate (CTR) prediction. CTR models suffer from a fundamental asymmetry:...

<https://arxiv.org/abs/2605.24989>

57 CONF

Interpretation, Learning, and Empathy as One Constraint: A Residual-Adequacy Architecture with Accountable Abstention

ArXiv CS.AI 50%

arXiv:2605.24999v1 Announce Type: cross Abstract: An agent must act on the situation before it, learn what it cannot yet represent, and model other agents well enough to coordinate. These faculties are usually realized by separate mechanisms, yet they share a failure mode: the...

<https://arxiv.org/abs/2605.24999>

58 CONF

Metropolis-Scale Resilient and Trustworthy Traffic Flow Inference Using Multi-Source Data

ArXiv CS.AI 50%

arXiv:2605.25004v1 Announce Type: cross Abstract: Inferring network-wide traffic states from sparse observations with high accuracy and trustworthy uncertainty quantification is essential for intelligent transportation systems, yet it remains challenging due to the...

<https://arxiv.org/abs/2605.25004>

59 CONF

Scale When Needed: Adaptive Neuron-level Mixed Precision Quantization Aware Training

ArXiv CS.AI 50%

arXiv:2605.25054v1 Announce Type: cross Abstract: Deploying deep neural networks on resource-constrained 6G edge devices demands aggressive compression with minimal accuracy loss. Quantization-Aware Training (QAT) has emerged as a leading compression approach; however, existing...

<https://arxiv.org/abs/2605.25054>

60 CONF

GL-LFGNN:A Global-Local Dual-branch Causal Graph Neural Network Based on Liang-Kleeman Information Flow for EEG Emotion Recognition

ArXiv CS.AI 50%

arXiv:2605.25061v1 Announce Type: cross Abstract: EEG-based emotion recognition holds significant promise for objective diagnosis of mood disorders. Graph neural networks (GNNs) have emerged as the dominant paradigm for modeling inter-channel dependencies in EEG, yet existing...

<https://arxiv.org/abs/2605.25061>

61 CONF

Polynomial Context-Truncation Sensitivity in Autoregressive Language Models: Sequential Wyner-Ziv Bounds for KV Cache Compression

ArXiv CS.AI 50%

arXiv:2605.25085v1 Announce Type: cross Abstract: We study the rate-distortion limits of online KV cache compression in autoregressive language models, formulating it as sequential Wyner-Ziv source coding on the filtration induced by the model, with the next-step query as...

<https://arxiv.org/abs/2605.25085>

62 CONF

Leveraging Gauge Freedom for Learning Non-Gradient Population Dynamics of Stochastic Systems

ArXiv CS.AI 50%

arXiv:2605.25107v1 Announce Type: cross Abstract: Existing work on population dynamics inference often focuses on flows arising from vector fields that are the gradients of scalar potentials. Among all admissible flows that are compatible with the population dynamics, gradient...

<https://arxiv.org/abs/2605.25107>

63 CONF

Uncertainty-DTW for Sequences and Visual Tokens

ArXiv CS.AI 50%

arXiv:2605.25110v1 Announce Type: cross Abstract: Aligning structured data is a fundamental problem in computer vision and machine learning, underlying tasks such as time series analysis, human action recognition, and visual representation learning. Existing alignment methods,...

<https://arxiv.org/abs/2605.25110>

64 CONF

Abduction-Deduction Entanglement: Domain Generalization via Representation Transplants

ArXiv CS.AI 50%

arXiv:2605.25156v1 Announce Type: cross Abstract: Prediction models trained under the source distribution do not generalize well to a different target distribution. A valid inference about an unseen data distribution must be anchored by the invariance of certain causal...

<https://arxiv.org/abs/2605.25156>

65 CONF

Influence-Inspired Spectral Rotations for Extreme Low-Bit LLM Quantization

ArXiv CS.AI 50%

arXiv:2605.25203v1 Announce Type: cross Abstract: We apply the influence-adaptive Walsh geometry of a companion theory paper (arXiv:2605.01637) to extreme low-bit weight-only LLM quantization. The recipe is one math-invariant transformation: WHT-rotate each linear layer's weight...

<https://arxiv.org/abs/2605.25203>

66 CONF

Multi-Objective Learning for Diffusion Models: A Statistical Theory under Semi-Supervised Learning

ArXiv CS.AI 50%

arXiv:2605.25210v1 Announce Type: cross Abstract: Diffusion models are increasingly used as powerful conditional generators, yet real deployments often involve multiple target distributions arising from different tasks, e.g., diverse prompt domains in text-to-image generation,...

<https://arxiv.org/abs/2605.25210>

67 CONF

Continuous-Depth Field Theory for Transformer Patching and Mechanistic Interpretability

ArXiv CS.AI 50%

arXiv:2605.25225v1 Announce Type: cross Abstract: Mechanistic interpretability often uses activation patching, causal tracing, path patching, and steering directions to reveal behaviorally meaningful directions in Transformer activation space. This paper develops a...

<https://arxiv.org/abs/2605.25225>

68 CONF

On the Epistemic Uncertainty of Overparametrized Neural Networks

ArXiv CS.AI 50%

arXiv:2605.25234v1 Announce Type: cross Abstract: Epistemic uncertainty is often viewed as a reducible uncertainty that vanishes with increasing data. This perspective implicitly assumes parameter identifiability and equates epistemic uncertainty with predictive variability. In...

<https://arxiv.org/abs/2605.25234>

69 CONF

Constraint-Anchored Attribution: Feasibility-Certified Counterfactuals and Bonferroni-PAC Sufficient Subsets for Neural CO Policies

ArXiv CS.AI 50%

arXiv:2605.25235v1 Announce Type: cross Abstract: We give an attribution method for neural combinatorial-optimisation (CO) policies that (i) decomposes a decision by constraint families via LP-relaxation duals, (ii) certifies counterfactuals through a combinatorial feasibility...

<https://arxiv.org/abs/2605.25235>

70 CONF

Positivity in classical enumerative geometry: a case study in synchronized AI-assisted mathematics

ArXiv CS.AI 50%

arXiv:2605.25271v1 Announce Type: cross Abstract: We study the symmetric polynomial $\prod_{i=1}^n (1 + \alpha_i x_i + \dots + \alpha_i x_n)$ where $A_{n,d} := \{\alpha \in \mathbb{Z}_{\geq 0}^n : |\alpha| = d\}$, which is the total Chern class of...

<https://arxiv.org/abs/2605.25271>

71 CONF

Neuromorphic LiDAR-based Bird's Eye View Object Detection using Energy-efficient Spiking Neural Networks

ArXiv CS.AI 50%

arXiv:2605.25293v1 Announce Type: cross Abstract: Autonomous driving perception demands accurate and efficient processing of three-dimensional sensor data under strict power constraints. Traditional convolutional neural networks achieve strong detection accuracy but are...

<https://arxiv.org/abs/2605.25293>

72 CONF

Certified Robustness from Approximate Gaussian Mixture Structures in Pretrained Latent Spaces

ArXiv CS.AI 50%

arXiv:2605.25352v1 Announce Type: cross Abstract: Deep learning models are vulnerable to adversarial perturbations, raising important concerns for safety-critical deployment. Empirical defenses can achieve strong robustness in practice, but lack formal guarantees, motivating the...

<https://arxiv.org/abs/2605.25352>

73 CONF

AI-Associated Lexical Shifts Across 34 Languages: Cross-Lingual Convergence and Diachronic Uptake in News Writing

ArXiv CS.AI 50%

arXiv:2605.25358v1 Announce Type: cross Abstract: AI-associated lexical shifts have been documented mainly in Scientific English. We extend this work to 34 languages in the WMT News Crawl corpus, refining a split-halves continuation diagnostic that compares GPT-4.1 continuations...

<https://arxiv.org/abs/2605.25358>

74 CONF

Anatomy-Anchored Self-Supervision: Distilling Vision Foundation Models for Invariant Ultrasound Representation

ArXiv CS.AI 50%

arXiv:2605.25402v1 Announce Type: cross Abstract: Self-supervised pre-training paradigm has gained increasing prominence for learning transferable representations in medical imaging, yet existing methods for ultrasound (US) images operate at the image or frame level, overlooking...

<https://arxiv.org/abs/2605.25402>

75 CONF

SeqRoute: Global Budget-Aware Sequential LLM Routing via Offline Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2605.25424v1 Announce Type: cross Abstract: Existing LLM routing frameworks treat queries as independent events, neglecting the sequential nature of real-world user sessions constrained by global computational budgets. This mismatch inevitably leads to budget bankruptcy:...

<https://arxiv.org/abs/2605.25424>

76 CONF

A Multi-Agent LLM Framework for Rating the Quality of Surgical Feedback

ArXiv CS.AI 50%

arXiv:2605.25440v1 Announce Type: cross Abstract: Verbal feedback delivered by attending surgeons in the operating room plays a critical formative role in resident trainee skill acquisition. Yet, assessing the quality of trainer feedback and its effectiveness in influencing...

<https://arxiv.org/abs/2605.25440>

77 CONF

Generative AI impacts on intra-urban inequality and skill premium in Beijing

ArXiv CS.AI 50%

arXiv:2605.25505v1 Announce Type: cross Abstract: Generative artificial intelligence (GenAI) is the first automation wave to reach high-cognitive tasks at scale, yet its effects on intra-urban inequality remain largely unknown. Using 5 million job postings from Beijing...

<https://arxiv.org/abs/2605.25505>

78 CONF

TopoAlign: Topology-Aware Visual Representation Alignment

ArXiv CS.AI 50%

arXiv:2605.25541v1 Announce Type: cross Abstract: Neural networks encode inputs as high-dimensional vectors, known as representations, that capture how models process data by encoding task-relevant structure and semantics. Representation alignment refers to the degree to which...

<https://arxiv.org/abs/2605.25541>

79 CONF

BC Protocol: Structured Dual-Expert Dialogue for Eliciting High-Quality Chain-of-Thought Post-Training Data

ArXiv CS.AI 50%

arXiv:2605.25549v1 Announce Type: cross Abstract: High-quality expert chain-of-thought (CoT) data is one of the core bottlenecks in large language model (LLM) post-training. Existing data production methods each have structural limitations: crowdsourced annotation lacks deep...

<https://arxiv.org/abs/2605.25549>

80 CONF

PennySynth: RAG-Driven Data Synthesis for Automated Quantum Code Generation

ArXiv CS.AI 50%

arXiv:2605.25572v1 Announce Type: cross Abstract: The growing complexity of quantum programming frameworks has exposed a critical limitation in existing large language model (LLM)-based code assistants: general-purpose models hallucinate PennyLane-specific gate names, misplace...

<https://arxiv.org/abs/2605.25572>

81 CONF

Fine-Tuning and Serving Gemma 4 31B on Google Cloud TPU: A Technical Comparison with GPU Baselines

ArXiv CS.AI 50%

arXiv:2605.25645v1 Announce Type: cross Abstract: We present the first end-to-end demonstration of fine-tuning and serving Google's Gemma 4 31B model on TPU hardware, providing an empirical comparison of TPU and GPU platforms for large language model adaptation. Using LoRA on a...

<https://arxiv.org/abs/2605.25645>

82 CONF

AutoSG: LLM-Driven Solver Generation Solely from Task Prompts for Expensive Optimization

ArXiv CS.AI 50%

arXiv:2605.25658v1 Announce Type: cross Abstract: Expensive optimization tasks are ubiquitous in real-world applications, demanding highly specialized solvers. While LLM-driven automated solver generation shows promise, current paradigms face three critical issues when tackling...

<https://arxiv.org/abs/2605.25658>

83 CONF

Meta-Engineering Harnesses for AI-Native Software Production: A Contract-Driven Adversarial Verification Architecture with Early Deployment Report

ArXiv CS.AI 50%

arXiv:2605.25665v1 Announce Type: cross Abstract: AI-native software development is often evaluated at the level of individual models, prompts, or generated artifacts. This framing is insufficient for production environments where software must be continuously produced,...

<https://arxiv.org/abs/2605.25665>

84 CONF

Referential Security as a New Paradigm for AI Evaluations

ArXiv CS.AI 50%

arXiv:2605.25673v1 Announce Type: cross Abstract: Security evaluations inherently depend on stable identifiers. Any finding, audit, or regulatory decision must remain attached to the specific artifact it pertains to. Continuously updated artificial intelligence systems violate...

<https://arxiv.org/abs/2605.25673>

85 CONF

Profiling-Driven Adaptive Distributed Transformer Inference on Embedded Edge Deployment

ArXiv CS.AI 50%

arXiv:2605.25682v1 Announce Type: cross Abstract: Distributing Transformer inference across embedded edge devices can alleviate individual memory and compute constraints, yet practical benefits on real hardware remain unclear: prior work relies largely on simulations that...

<https://arxiv.org/abs/2605.25682>

86 CONF

How Should LLMs Consume High-Quality Data? Optimal Data Scheduling via Quality-Aware Functional Scaling Laws

ArXiv CS.AI 50%

arXiv:2605.25698v1 Announce Type: cross Abstract: High-quality data is scarce in large language model (LLM) training, yet how to schedule its use jointly with training dynamics lacks theoretical guidance. We extend functional scaling laws by incorporating a data-quality...

<https://arxiv.org/abs/2605.25698>

87 CONF

MDGMIX: Boundary-Aware Subgraph Mixing for Multi-Domain Graph Pre-Training

ArXiv CS.AI 50%

arXiv:2605.25771v1 Announce Type: cross Abstract: Multi-domain graph pre-training is a crucial step in constructing foundational graph models with cross-domain generalization capabilities. However, existing methods predominantly rely on jointly training all source domain graphs,...

<https://arxiv.org/abs/2605.25771>

88 CONF

SAMark: A Self-Anchored Text Watermarking with Paragraph-Level Paraphrase Robustness

ArXiv CS.AI 50%

arXiv:2605.25796v1 Announce Type: cross Abstract: Semantic-level watermarking (SWM) improves robustness against text modifications by treating sentences as the basic unit. However, robustness to paragraph-level paraphrasing remains difficult because such attacks globally disrupt...

<https://arxiv.org/abs/2605.25796>

89 CONF

Adaptive Graph Refinement and Label Propagation with LLMs for Cost-Effective Entity Resolution

ArXiv CS.AI 50%

arXiv:2605.25814v1 Announce Type: cross Abstract: Dirty entity resolution (ER), which identifies records referring to the same real-world entity from a single, messy dataset, is a fundamental task in data management and mining. However, the dominant blocking-matching-clustering...

<https://arxiv.org/abs/2605.25814>

90 CONF

Fine-Tuning Over Architectural Complexity: Broad-Coverage PII Detection on PIIBench with DeBERTa

ArXiv CS.AI 50%

arXiv:2605.25816v1 Announce Type: cross Abstract: Personally identifiable information (PII) detection systems are frequently trained within narrow source or domain boundaries, limiting coverage when deployed on heterogeneous text. We study model fine-tuning on a corrected...

<https://arxiv.org/abs/2605.25816>

91 CONF

Dynamic Relational Priming Improves Transformer in Multivariate Time Series

ArXiv CS.AI 50%

arXiv:2509.12196v2 Announce Type: replace-cross Abstract: Standard attention mechanisms in transformers employ static token representations that remain unchanged across all pair-wise computations in each layer. This limits their representational alignment with the potentially...

<https://arxiv.org/abs/2509.12196>

92 CONF

TIAR: Trajectory-Informed Advantage Reweighting for LLM Abstention Learning

ArXiv CS.AI 50%

arXiv:2605.25850v1 Announce Type: cross Abstract: This paper investigates large language model (LLM) abstention learning, specifically using ternary reward, which incentivize truthfulness in large language models. This paper extends that idea by moving from a ternary reward to a...

<https://arxiv.org/abs/2605.25850>

93 CONF

Explaining Too Much? Understanding How Large Language Model Reasoning Traces Influence Performance and Metacognition

ArXiv CS.AI 50%

arXiv:2605.25856v1 Announce Type: cross Abstract: Large Language Model interfaces are increasingly verbose, exposing intermediate reasoning traces alongside final answers. Traces are framed as transparency mechanisms, yet it is unclear how people use them to solve problems. We...

<https://arxiv.org/abs/2605.25856>

94 CONF

Causal Tongue-Tie: LLMs Can Encode Causal Direction, But Their Yes/No Outputs Fail to Express

ArXiv CS.AI 50%

arXiv:2605.25891v1 Announce Type: cross Abstract: We find a mismatch between what large language models encode about a causal question and what they answer. On anti-commonsense CLadder items, a fixed linear probe recovers the evidence-supported answer from the model's hidden...

<https://arxiv.org/abs/2605.25891>

95 CONF

EchoPilot: Training-Free Ultrasound Video Segmentation via Scale-Space Semantic Prompting and Reliability-Gated Memory

ArXiv CS.AI 50%

arXiv:2605.25944v1 Announce Type: cross Abstract: Ultrasound video segmentation is clinically valuable yet difficult due to speckle noise, weak boundaries, and rapid anatomical deformation. Recent promptable foundation models enable point-guided segmentation, but their direct...

<https://arxiv.org/abs/2605.25944>

96 CONF

Creative Quality Alignment: Expert Tacit Knowledge Transfer via Chain-of-Thought Fine-Tuning

ArXiv CS.AI 50%

arXiv:2605.25977v1 Announce Type: cross Abstract: This paper provides an empirical implementation of the creative quality metric proposed in Calibrated Surprise (Zou & Xu, 2026a). The question this paper addresses is: does this mathematical claim hold at the engineering level?...

<https://arxiv.org/abs/2605.25977>

97 CONF

SafeCtrl-RL: Inference-Time Adaptive Behaviour Control for LLM Dialogue via RL-Driven Prompt Optimisation

ArXiv CS.AI 50%

arXiv:2605.25984v1 Announce Type: cross Abstract: Ensuring safe and contextually appropriate behaviour in Large Language Models (LLMs) remains a critical challenge for real-world deployment. We present \textbf{SafeCtrl-RL}, an inference-time behavioural control framework that...

<https://arxiv.org/abs/2605.25984>

98 CONF

AI-Assisted Systematization for Evaluating GenAI Systems

ArXiv CS.AI 50%

arXiv:2605.26001v1 Announce Type: cross Abstract: Evaluating generative AI (GenAI) systems is challenging because many targets of evaluation are broad, contested concepts, such as "reasoning," "fairness," or "creativity." When these concepts are left underspecified, it becomes...

<https://arxiv.org/abs/2605.26001>

99 **CONF****Learning in Low-Dimensional Subspaces: Orthogonal Bottlenecks for Reinforcement Learning**

ArXiv CS.AI 50%

arXiv:2605.26012v1 Announce Type: cross Abstract: Deep reinforcement learning (RL) agents commonly rely on high-dimensional neural representations, despite growing evidence that task-relevant value and policy structure may be intrinsically low-dimensional. In this work, we...

<https://arxiv.org/abs/2605.26012>

00 **CONF****Confidence and Calibration of Activation Oracles for Reliable Interpretation of Language Model Internals**

ArXiv CS.AI 50%

arXiv:2605.26045v1 Announce Type: cross Abstract: Activation oracles aim to make the activations of other models legible to humans and yield promising results compared to white-box interpretability techniques. However, uncertainty quantification (UQ) for the natural-language...

<https://arxiv.org/abs/2605.26045>

01 **CONF****When Gradients Collide: Failure Modes of Multi-Objective Prompt Optimization for LLM Judges**

ArXiv CS.AI 50%

arXiv:2605.26046v1 Announce Type: cross Abstract: Customizing an LLM judge to a specific task or domain often involves optimizing its prompt across multiple evaluation criteria simultaneously. Textual gradient methods automate this for a single judge criterion, however they...

<https://arxiv.org/abs/2605.26046>

02 **CONF****Channel-wise Vector Quantization**

ArXiv CS.AI 50%

arXiv:2605.26089v1 Announce Type: cross Abstract: We present Channel-wise Vector Quantization (CVQ), a novel image tokenization paradigm that replaces patch-wise tokens with channel-wise tokens. Unlike conventional vector quantization, which assigns a discrete token to each...

<https://arxiv.org/abs/2605.26089>

03 **CONF****Language Models Need Sleep**

ArXiv CS.AI 50%

arXiv:2605.26099v1 Announce Type: cross Abstract: Transformer-based large language models are increasingly used for long-horizon tasks; however, their attention mechanism scales poorly with context length. To handle this, we study a sleep-like consolidation mechanism in which a...

<https://arxiv.org/abs/2605.26099>

04 **CONF****Probing the Preferences of a Language Model: Integrating Verbal and Behavioral Tests of AI Welfare**

ArXiv CS.AI 50%

arXiv:2509.07961v2 Announce Type: replace Abstract: We develop new experimental paradigms for measuring welfare in language models. We compare verbal reports of models about their preferences with preferences expressed through behavior when navigating a virtual environment and...

<https://arxiv.org/abs/2509.07961>

05 CONF

False Fixed Points: Kantian Feedback, Stable Miscalibration, and Representational Compression in LLMs

ArXiv CS.AI 50%

arXiv:2510.14925v4 Announce Type: replace Abstract: High-confidence errors in large language models are often treated as fragile failures. We study an alternative: some errors may be false fixed points, locally stable, internally coherent, and confidently wrong. This separates...

<https://arxiv.org/abs/2510.14925>

06 CONF

From Prompt Optimization to Multi-Dimensional Credibility Evaluation: Enhancing Trustworthiness of Chinese LLM-Generated Liver MRI Reports -- with Preliminary Extension to Lung Cancer

ArXiv CS.AI 50%

arXiv:2510.23008v3 Announce Type: replace Abstract: Large language models (LLMs) have demonstrated promising performance in generating diagnostic conclusions from imaging findings, thereby supporting radiology reporting, trainee education, and quality control. However,...

<https://arxiv.org/abs/2510.23008>

07 CONF

Chain-of-Thought Hijacking

ArXiv CS.AI 50%

arXiv:2510.26418v4 Announce Type: replace Abstract: Large Reasoning Models (LRMs) improve task performance through extended inference-time reasoning. Although previous studies suggest that longer reasoning should lead to more robust safety behavior, we find evidence to the...

<https://arxiv.org/abs/2510.26418>

08 CONF

Optimizing Sensor Placement for Flow Reconstruction in Urban Drainage Networks: A Digital Twin-Based Sparse Sensing Approach

ArXiv CS.AI 50%

arXiv:2511.04556v2 Announce Type: replace Abstract: Urban flooding triggered by intense rainfall is becoming increasingly frequent and widespread. While flood prediction and monitoring in high spatio-temporal resolution are desired, practical constraints in time, budget, and...

<https://arxiv.org/abs/2511.04556>

09 CONF

Actionable and diverse counterfactual explanations incorporating domain knowledge and plausibility constraints

ArXiv CS.AI 50%

arXiv:2511.20236v3 Announce Type: replace Abstract: Counterfactual explanations improve the actionable interpretability of machine learning models by identifying minimal changes required to achieve a desired outcome. However, existing methods often neglect dependencies among...

<https://arxiv.org/abs/2511.20236>

10 CONF

PathWise: Planning through World Model for Automated Heuristic Design via Self-Evolving LLMs

ArXiv CS.AI 50%

arXiv:2601.20539v3 Announce Type: replace Abstract: Large Language Models (LLMs) have enabled automated heuristic design (AHD) for combinatorial optimization problems (COPs), but existing frameworks' reliance on fixed evolutionary rules and static prompt templates often leads to...

<https://arxiv.org/abs/2601.20539>

11 CONF

Why Your Deep Research Agent Fails? On Hallucination Evaluation in Full Research Trajectory

ArXiv CS.AI 50%

arXiv:2601.22984v2 Announce Type: replace Abstract: Diagnosing failure patterns in Deep Research Agents (DRAs) remains a critical challenge. Existing benchmarks predominantly rely on end-to-end evaluation, obscuring intermediate hallucinations that accumulate throughout the...

<https://arxiv.org/abs/2601.22984>

12 CONF

MedBeads: An Agent-Native, Immutable Data Substrate for Trustworthy Medical AI

ArXiv CS.AI 50%

arXiv:2602.01086v2 Announce Type: replace Abstract: Background: As of 2026, Large Language Models (LLMs) demonstrate expert-level medical knowledge. However, deploying them as autonomous "Clinical Agents" remains limited. Current Electronic Medical Records (EMRs) and standards...

<https://arxiv.org/abs/2602.01086>

13 CONF

FLINGO -- Instilling ASP Expressiveness into Linear Integer Constraints

ArXiv CS.AI 50%

arXiv:2602.09620v2 Announce Type: replace Abstract: Constraint Answer Set Programming (CASP) is a hybrid paradigm that enriches Answer Set Programming (ASP) with numerical constraint processing, a crucial requirement for many real-world applications. However, the specification...

<https://arxiv.org/abs/2602.09620>

14 CONF

JEPA-DNA: Grounding Genomic Foundation Models through Joint-Embedding Predictive Architectures

ArXiv CS.AI 50%

arXiv:2602.17162v2 Announce Type: replace Abstract: Genomic Foundation Models (GFM) typically rely on Masked Language Modeling (MLM) or Next-Token Prediction (NTP) to learn the "Laws of Nature". While effective at capturing local syntax, these generative paradigms prioritize...

<https://arxiv.org/abs/2602.17162>

15 CONF

All Leaks Count, Some Count More: Interpretable Temporal Contamination Detection and Mitigation in LLM Backtesting

ArXiv CS.AI 50%

arXiv:2602.17234v2 Announce Type: replace Abstract: Backtesting LLMs on resolved events assumes models reason only from pre-cutoff knowledge, yet pretrained models inevitably leak post-cutoff knowledge. We introduce a claim-level evaluation framework that decomposes prediction...

<https://arxiv.org/abs/2602.17234>

16 CONF

Is VLA Reasoning Faithful? Probing Safety of Chain-of-Causation in Autonomous Driving Models

ArXiv CS.AI 50%

arXiv:2605.17268v2 Announce Type: replace Abstract: We present the first systematic study of faithfulness in Vision-Language-Action (VLA) driving models, analyzing 300 Alpamayo-R1-10B inferences across 100 diverse PhysicalAI-AV scenarios. Our main finding is that output...

<https://arxiv.org/abs/2605.17268>

17 CONF

ECUAS_n\$: A family of metrics for principled evaluation of uncertainty-augmented systems

ArXiv CS.AI 50%

arXiv:2605.20490v3 Announce Type: replace Abstract: In high-stakes automated decision-making, access to predictive uncertainty is essential for enabling users -- human or downstream systems -- to accept or reject predictions based on application-specific cost trade-offs. Such...

<https://arxiv.org/abs/2605.20490>

18 CONF

SMDD-Bench: Can LLMs Solve Real-World Small Molecule Drug Design Tasks?

ArXiv CS.AI 50%

arXiv:2605.21740v2 Announce Type: replace Abstract: LLM agents have incredible potential for scientific discovery applications. However, the performance of LLM agents on real-world, small molecule drug design (SMDD) tasks across diverse chemistries and targets is unclear....

<https://arxiv.org/abs/2605.21740>

19 CONF

Meta-Soft: Leveraging Composable Meta-Tokens for Context-Preserving KV Cache Compression

ArXiv CS.AI 50%

arXiv:2605.22337v2 Announce Type: replace Abstract: The KV cache used in large language models has linearly growing time complexity, so LLMs face memory blow-up and reduced decoding efficiency when they process long contexts. Current KV Cache eviction has become an important...

<https://arxiv.org/abs/2605.22337>

20 CONF

AMEL: Accumulated Message Effects on LLM Judgments

ArXiv CS.AI 50%

arXiv:2605.22714v2 Announce Type: replace Abstract: Large language models are routinely used as automated evaluators: to review code, moderate content, or score outputs, often with many items passing through one conversation. We ask whether the polarity of prior conversation...

<https://arxiv.org/abs/2605.22714>

21 CONF

The Meme Is the Message: Generative Memes and AI Visuals in the 2024 USA Presidential Elections

ArXiv CS.AI 50%

arXiv:2411.00934v2 Announce Type: replace-cross Abstract: Visual content on social media has become increasingly influential in shaping political discourse and civic engagement, but it also limits participation due to the increased cost of multimedia production. In tandem, the...

<https://arxiv.org/abs/2411.00934>

22 CONF

Fusion Embedding for Pose-Guided Person Image Synthesis with Diffusion Model

ArXiv CS.AI 50%

arXiv:2412.07333v2 Announce Type: replace-cross Abstract: Pose-Guided Person Image Synthesis (PGPIS) aims to generate human images in specified poses while preserving the identity and appearance of a source image. This technology facilitates diverse applications, including...

<https://arxiv.org/abs/2412.07333>

23 CONF

Message-Passing GNNs Fail to Approximate Sparse Triangular Factorizations

ArXiv CS.AI 50%

arXiv:2502.01397v3 Announce Type: replace-cross Abstract: Graph Neural Networks (GNNs) have been proposed as a tool for learning sparse matrix preconditioners, which are key components in accelerating linear solvers. We present theoretical and empirical evidence that...

<https://arxiv.org/abs/2502.01397>

24 CONF

ExplainReduce: Generating global explanations from many local explanations

ArXiv CS.AI 50%

arXiv:2502.10311v3 Announce Type: replace-cross Abstract: Most commonly used non-linear machine learning methods are closed-box models, uninterpretable to humans. The field of explainable artificial intelligence (XAI) aims to develop tools to examine the inner workings of these...

<https://arxiv.org/abs/2502.10311>

25 CONF

Pragmatic Reasoning improves LLM Code Generation

ArXiv CS.AI 50%

arXiv:2502.15835v5 Announce Type: replace-cross Abstract: Pragmatic reasoning helps interlocutors infer intended meaning from ambiguous or underspecified messages by considering shared context and counterfactual alternatives. Similar challenges arise in natural language-to-code...

<https://arxiv.org/abs/2502.15835>

26 CONF

How does Bayesian Sampling help Membership Inference Attacks?

ArXiv CS.AI 50%

arXiv:2503.07482v2 Announce Type: replace-cross Abstract: Membership Inference Attacks (MIAs) aim to estimate whether a specific data point was used in the training of a given model. Existing state-of-the-art attacks typically rely on training multiple reference models to...

<https://arxiv.org/abs/2503.07482>

27 CONF

SoK: A Comprehensive Security Analysis of Jailbreak Resilience in GPT and DeepSeek Models

ArXiv CS.AI 50%

arXiv:2506.18543v2 Announce Type: replace-cross Abstract: The rapid proliferation of Large Language Models (LLMs) has heightened concerns regarding their exposure to jailbreak attacks, which craft adversarial inputs designed to elicit unsafe content. Although proprietary models...

<https://arxiv.org/abs/2506.18543>

28 CONF

Designing Singing Syllabi with Virtual Avatars: AI-Assisted Syllabus Reauthoring

ArXiv CS.AI 50%

arXiv:2508.11872v3 Announce Type: replace-cross Abstract: Traditional syllabi often function as static reference documents rather than engaging introductions to a course. In practical teaching, we observe that few students thoroughly read or fully comprehend the information...

<https://arxiv.org/abs/2508.11872>

29 CONF

KAME: Tandem Architecture for Enhancing Knowledge in Real-Time Speech-to-Speech Conversational AI

ArXiv CS.AI 50%

arXiv:2510.02327v2 Announce Type: replace-cross Abstract: Real-time speech-to-speech (S2S) models excel at generating natural, low-latency conversational responses but often lack deep knowledge and semantic understanding. Conversely, cascaded systems combining automatic speech...

<https://arxiv.org/abs/2510.02327>

30 CONF

Safety in Embodied AI: A Survey of Risks, Attacks, and Defenses

ArXiv CS.AI 50%

arXiv:2605.02900v2 Announce Type: replace-cross Abstract: Embodied Artificial Intelligence (Embodied AI) integrates perception, cognition, planning, and interaction into agents that operate in open-world, safety-critical environments. As these systems gain autonomy and enter...

<https://arxiv.org/abs/2605.02900>

31 CONF

AI-generated podcasts: Synthetic Intimacy and Cultural Mistranslation in NotebookLM's Audio Overviews

ArXiv CS.AI 50%

arXiv:2511.08654v2 Announce Type: replace-cross Abstract: This paper analyses AI-generated podcasts produced by Google's NotebookLM, which generates audio podcasts with two chatty AI hosts discussing whichever documents a user uploads. While AI-generated podcasts have been...

<https://arxiv.org/abs/2511.08654>

32 CONF

Bridging the Semantic-Action Gap in Visual Token Pruning for Efficient VLA Inference

ArXiv CS.AI 50%

arXiv:2511.16449v4 Announce Type: replace-cross Abstract: Vision-Language-Action (VLA) models have shown great potential for embodied AI by integrating visual perception, language understanding, and action execution. In real-time deployment, these models must process continuous...

<https://arxiv.org/abs/2511.16449>

33 CONF

Understanding, Accelerating, and Improving MeanFlow Training

ArXiv CS.AI 50%

arXiv:2511.19065v2 Announce Type: replace-cross Abstract: MeanFlow promises high-quality generative modeling in few steps, by jointly learning instantaneous and average velocity fields. Yet, the underlying training dynamics remain unclear. We analyze the interaction between the...

<https://arxiv.org/abs/2511.19065>

34 CONF

Asking LLMs to Verify First is Almost Free Lunch

ArXiv CS.AI 50%

arXiv:2511.21734v2 Announce Type: replace-cross Abstract: To enhance the reasoning capabilities of Large Language Models (LLMs) without high costs of training, nor extensive test-time sampling, we introduce Verification-First (VF), a strategy that prompts models to verify a...

<https://arxiv.org/abs/2511.21734>

35 CONF

AI as Equalizer or Amplifier? Task Complexity as the Moderating Factor for Human Expertise in Hybrid Intelligence Systems

ArXiv CS.AI 50%

arXiv:2512.10961v2 Announce Type: replace-cross Abstract: A growing body of empirical research suggests that generative AI narrows performance gaps between novice and expert workers on routine tasks--the so-called "equalizer" effect. This paper challenges the generality of that...

<https://arxiv.org/abs/2512.10961>

36 CONF

\$M^3-Verse\$: A "Spot the Difference" Challenge for Large Multimodal Models

ArXiv CS.AI 50%

arXiv:2512.18735v2 Announce Type: replace-cross Abstract: Modern Large Multimodal Models (LMMs) have demonstrated extraordinary ability in static image and single-state spatial-temporal understanding. However, their capacity to comprehend the dynamic changes of objects within a...

<https://arxiv.org/abs/2512.18735>

37 CONF

QASA: Quality-Aware Semantic Augmentation for Robust Multimodal Sentiment Analysis

ArXiv CS.AI 50%

arXiv:2601.06870v2 Announce Type: replace-cross Abstract: Multimodal large language models have demonstrated strong ability in capturing semantic representations for multimodal sentiment analysis. Their capacity to learn stable and generalizable multimodal features is limited,...

<https://arxiv.org/abs/2601.06870>

38 CONF

Unifying Speech Editing Detection and Content Localization via Prior-Enhanced Audio LLMs

ArXiv CS.AI 50%

arXiv:2601.21463v3 Announce Type: replace-cross Abstract: Existing speech editing detection (SED) datasets are predominantly constructed using manual splicing or limited editing operations, resulting in restricted diversity and poor coverage of realistic editing scenarios....

<https://arxiv.org/abs/2601.21463>

39 CONF

RecGOAT: Graph Optimal Adaptive Transport for LLM-Enhanced Multimodal Recommendation with Dual Semantic Alignment

ArXiv CS.AI 50%

arXiv:2602.00682v2 Announce Type: replace-cross Abstract: Integrating large language model (LLM) representations into multimodal recommendation has shown promise, yet a fundamental challenge remains largely overlooked: the semantic heterogeneity between generative LM...

<https://arxiv.org/abs/2602.00682>

40 CONF

Generative Visual Code Mobile World Models

ArXiv CS.AI 50%

arXiv:2602.01576v2 Announce Type: replace-cross Abstract: Mobile Graphical User Interface (GUI) World Models (WMs) offer a promising path for improving mobile GUI agent performance at train- and inference-time. However, current approaches face a critical trade-off: text-based...

<https://arxiv.org/abs/2602.01576>

41 CONF

Reward-free Alignment for Conflicting Objectives

ArXiv CS.AI 50%

arXiv:2602.02495v3 Announce Type: replace-cross Abstract: Direct alignment methods are increasingly used to align large language models (LLMs) with human preferences. However, many real-world alignment problems involve multiple conflicting objectives, where naive aggregation of...

<https://arxiv.org/abs/2602.02495>

42 CONF

Scalable Explainability-as-a-Service (XaaS) for Edge AI Systems

ArXiv CS.AI 50%

arXiv:2602.04120v3 Announce Type: replace-cross Abstract: Though Explainable AI (XAI) has made significant advancements, its inclusion in edge and IoT systems is typically ad-hoc and inefficient. Most current methods are "coupled" in such a way that they generate explanations...

<https://arxiv.org/abs/2602.04120>

43 CONF

Red-Teaming Claude Opus and ChatGPT-based Security Advisors for Trusted Execution Environments

ArXiv CS.AI 50%

arXiv:2602.19450v2 Announce Type: replace-cross Abstract: Trusted Execution Environments (TEEs) (e.g., Intel SGX and ArmTrustZone) aim to protect sensitive computation from a compromised operating system, yet real deployments remain vulnerable to microarchitectural leakage,...

<https://arxiv.org/abs/2602.19450>

44 CONF

MoBiQuant: Mixture-of-Bits Quantization for Token-Adaptive Any-Precision LLM

ArXiv CS.AI 50%

arXiv:2602.20191v2 Announce Type: replace-cross Abstract: Dynamic runtime latency and memory constraints necessitate flexible large language model (LLM) deployment, where an LLM can be inferred with various quantization precisions based on available computational resources....

<https://arxiv.org/abs/2602.20191>

45 CONF

Learning from Trials and Errors: Reflective Test-Time Planning for Embodied LLMs

ArXiv CS.AI 50%

arXiv:2602.21198v3 Announce Type: replace-cross Abstract: Embodied LLMs endow robots with high-level task reasoning, but they cannot reflect on what went wrong or why, turning deployment into a sequence of independent trials where mistakes repeat rather than accumulate into...

<https://arxiv.org/abs/2602.21198>

46 CONF

MultiPUFFIN: A Multimodal Domain-Constrained Foundation Model for Molecular Property Prediction of Small Molecules

ArXiv CS.AI 50%

arXiv:2603.00857v2 Announce Type: replace-cross Abstract: MultiPUFFIN is a domain-informed multimodal foundation model for predicting thermophysical properties of small molecules, addressing a critical gap in chemical engineering, drug discovery, and materials science. Existing...

<https://arxiv.org/abs/2603.00857>

47 CONF

Discounted Beta-Bernoulli Reward Estimation for Sample-Efficient Reinforcement Learning with Verifiable Rewards

ArXiv CS.AI 50%

arXiv:2603.18444v2 Announce Type: replace-cross Abstract: Reinforcement learning with verifiable rewards (RLVR) has emerged as an effective post-training paradigm for improving the reasoning capabilities of large language models. However, existing group-based RLVR methods often...

<https://arxiv.org/abs/2603.18444>

48 CONF

Rethinking the Comparison Unit in Sequence-Level Reinforcement Learning: An Equal-Length Paired Training Framework from Loss Correction to Sample Construction

ArXiv CS.AI 50%

arXiv:2604.17328v2 Announce Type: replace-cross Abstract: This paper investigates the length problem in sequence-level relative reinforcement learning. We observe that, although existing methods partially alleviate length-related phenomena, a more fundamental issue remains...

<https://arxiv.org/abs/2604.17328>

49 CONF

Copy-as-Decode: Grammar-Constrained Parallel Prefill for LLM Editing

ArXiv CS.AI 50%

arXiv:2604.18170v2 Announce Type: replace-cross Abstract: LLMs edit text and code by autoregressively regenerating the full output, even when most tokens appear verbatim in the input. We study Copy-as-Decode, a decoding-layer mechanism that recasts edit generation as structured...

<https://arxiv.org/abs/2604.18170>

50 CONF

Chain of Evidence: Pixel-Level Visual Attribution for Iterative Retrieval-Augmented Generation

ArXiv CS.AI 50%

arXiv:2605.01284v2 Announce Type: replace-cross Abstract: Iterative Retrieval-Augmented Generation (iRAG) has emerged as a powerful paradigm for answering complex multi-hop questions by progressively retrieving and reasoning over external documents. However, current systems...

<https://arxiv.org/abs/2605.01284>

51 CONF

Auditing Stealth Sycophancy in Mental-Health Dialogue: Structured Clinical-State Diagnostics and Clean Matched Benchmarks

ArXiv CS.AI 50%

arXiv:2605.03472v2 Announce Type: replace-cross Abstract: Mental-health dialogue models are increasingly evaluated by AI-based evaluators, yet these evaluators often treat surface empathy, supportiveness, or fluency as evidence of safety. In this paper, we study a hidden failure...

<https://arxiv.org/abs/2605.03472>

52 CONF

LLMs Uncertainty Quantification via Adaptive Conformal Semantic Entropy

ArXiv CS.AI 50%

arXiv:2605.04295v2 Announce Type: replace-cross Abstract: LLMs' overconfidence, particularly when hallucinating, poses a significant challenge for the deployment of the models in safety-critical settings and makes a reliable estimation of uncertainty necessary. Existing...

<https://arxiv.org/abs/2605.04295>

53 CONF

PASA: A Principled Embedding-Space Watermarking Approach for LLM-Generated Text under Semantic-Invariant Attacks

ArXiv CS.AI 50%

arXiv:2605.10977v2 Announce Type: replace-cross Abstract: Watermarking for large language models (LLMs) is a promising approach for detecting LLM-generated text and enabling responsible deployment. However, existing watermarking methods are often vulnerable to semantic-invariant...

<https://arxiv.org/abs/2605.10977>

54 CONF

Persona-Model Collapse in Emergent Misalignment

ArXiv CS.AI 50%

arXiv:2605.12850v2 Announce Type: replace-cross Abstract: Fine-tuning large language models on narrow data with harmful content produces broadly misaligned behavior on unrelated prompts, a phenomenon known as emergent misalignment. We propose that emergent misalignment involves...

<https://arxiv.org/abs/2605.12850>

55 CONF

Position: State-of-the-Art Claims Require State-of-the-Art Evidence

ArXiv CS.AI 50%

arXiv:2605.17273v2 Announce Type: replace-cross Abstract: State-of-the-Art (SOTA) claims pervade Artificial Intelligence (AI) and Machine Learning (ML) research. These claims rest on benchmark evaluations, where models are ranked by aggregate scores across tasks. Public...

<https://arxiv.org/abs/2605.17273>

56 CONF

ProcCtrlBench: Evaluating Process-Level Defects and Control Preservation in LLM Coding Agents

ArXiv CS.AI 50%

arXiv:2605.20251v3 Announce Type: replace-cross Abstract: Existing benchmarks for LLM coding agents primarily evaluate final outcomes. While useful for measuring overall capability, these metrics provide limited visibility and often miss defects that arise during execution. We...

<https://arxiv.org/abs/2605.20251>

57 CONF

TimeGuard: Channel-wise Pool Training for Backdoor Defense in Time Series Forecasting

ArXiv CS.AI 50%

arXiv:2605.22365v2 Announce Type: replace-cross Abstract: Time Series Forecasting (TSF) is highly vulnerable to backdoor attacks, yet effective defenses remain underexplored due to challenges arising from data entanglement and shifts in task formulation. To fill this gap, we...

<https://arxiv.org/abs/2605.22365>

58 CONF

Contractual Skills: A GovernSpec Design Framework for Enterprise AI Agents

ArXiv CS.AI 50%

arXiv:2605.22634v2 Announce Type: replace-cross Abstract: Skills have become a practical packaging mechanism for agent instructions, workflows, scripts, and reference materials. In enterprise settings, however, a skill often needs to express more than task guidance: goals, input...

<https://arxiv.org/abs/2605.22634>